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OSCALC – Opening Shock Calculator

Version 1.01

User's manual

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Acknowledgment

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Version Tracker

This manual covers the use of both OSCALC V1.0 and V1.01. Version 1.01 is the same as version of V1.0, except for slightly modified input box titles.

Warning - Disclaimer

OSCALC provides the means to estimate the value of the maximum drag generated during parachute inflation, based on inputs provided by the user. The authors and their governmental funding agency cannot make any claim on the accuracy of the results.

1. What is OSCALC?

1. Opening Shock CALCulator† – What is it?

- **It is a simple program that estimates the maximum (drag) force F_{max} generated during parachute inflation**
- **Uses inputs that are "straightforward" to obtain**
- **Calculation applies to *any* parachute design and reefing type:**
  - **low- and high-porosity hemisphericals (unreefed, reefed, dis-reefing)**
  - **parafoils (unreefed, line-reefed, slider-reefed)**
  - **in fact, *anything that can be used as a parachute***
- **Is it based on an equation commonly used in parachute engineering:**

OSCALC computes this number

From graph in pop-under/over window

$$F_{\max} = \left(\frac{1}{2}\rho V_{stretch}^{2}\right)\left(SC_D\right)_{sd} C_k$$

From the inputs of the problem

---

†**Pronounced as *"O"-"S" – "CALC"***

OSCALC – other features; restrictions

Features

- Run on the Microsoft Windows operating system
- Inputs can be made using any unit system
- Instant processing speed

Restrictions

- Restricted to parachutes attached to unpowered payloads (no motors – gravity is OK); there is no guarantee that OSCALC will produce, for example, an accurate calculation of the maximum force sustained by a parachute deploying while being connected to an ejection seat propelled by a rocket
- OSCALC is not a true predictor of opening shock since it requires the use of t_{fill} (n_{fill}), which is an actual inflation performance variable; but there is a lot of n_{fill} -data in the public domain already; otherwise measuring t_{fill} from video is a straightforward task

What applications is it most useful for?

- **F_{max}-estimation at the DZ, right after a test: OSCALC provides the means for calculating F_{max}, based solely on the basic canopy dimensions & drag properties, payload weight, deployment conditions and video of the inflation process (for inflation time info). The tool should be very useful during those tests where the payload is not equipped with load cells or accelerometers.**

- **Provides a good guess for F_{max}, even for new parachute and reefing designs, including designs that have not yet been documented in the public domain (see Section 6.6 for suggestions)**

What applications is it most useful for? – cont'd

- **"Sanity"-check for developers of computer simulations of the inflation process**

- **Calculation of F_{max} sustained during inflation scenarios that are not covered by computer simulations of inflation such as PIMS, FSI/CFD, Sandia model, etc.; for example: malfunctions, mis-staged openings, etc. Again, all one needs here is video to get the inflation time information**

What is the unit system of OSCALC?

- **No physical constants are being used explicitly – the only dimensional equation is** $F_{\max} = \left(\frac{1}{2}\rho V_{stretch}^{2}\right)\left(SC_D\right)_{sd} C_k$

- **Most variables are used in non-dimensional form (ratios)**

Input values can be entered in any units, as long as they are balanced and consistent. We recommend using either:

- **American Standard Units, based on *feet*, *seconds*, *slugs and pounds***

or

- **Metric Units, based on *meters, seconds, kilograms and Newtons***

The examples of Section 6 show how to enter OSCALC inputs using American Standard Units.

2. Basic input data

2. Basic input data

$$F_{\max} = \left(\frac{1}{2}\rho V_{stretch}^{2}\right)\left(SC_D\right)_{sd} C_k$$

- $V_{stretch}$ = parachute-payload fall rate at line stretch; or at the moment of "first air" for canopy-first deployments; or whenever the canopy is stretched and its mouth opened (or wing inlets opened)

- ρ = Atmopsheric density at deployment altitude

- $(SC_D)_{sd}$ = Drag area of the fully opened canopy during steady descent.
  - For hemispherical canopies $(SC_D)_{sd} = C_{D0}\, S_0 = C_{D0}\, (\pi\, D_0^2/4)$ with S_0 being to total canopy area, including vents' area [1]
  - For parafoils $(SC_D)_{sd} = C_{D0}\, S_0 \sim 1.0 \times$ *wing chord* $\times$ *wing span*
  - For a cluster of hemispherical canopies - see example 6.5 in Section 6 below

- C_k = Opening shock factor - What are the values of C_k?

Basic references about the Opening Shock Factor C_k

[1] T. W. Knacke, "Parachute Recovery Systems Design Manual"; Para Publishing (Santa Barbara, CA 1992).

[2] Ewing, E. G., Bixby, H. W. and Knacke, T. W.; "Recovery Systems Design guide"; pp. 254 – 257; report AFFL-TR-78-151. Submitted to: Air Force Flight Dynamics Laboratory, AF Wright Aeronautical Laboratories, Wright-Patterson Air Forced Base, December 1978. Unpublished.

[3] Wolf, D., "Opening Shock", AIAA-99-1702, 15th CEAS/AIAA Aerodynamic Decelerator Systems Technology Conference, Toulouse, France, 8-11 June 1999.

[4] Potvin, J.; "On Opening Shock Factor - Mass Ratio Universality"; To appear in *Journal of Aircraft* (2006) Copies of manuscript available on request.

[5] Potvin, J., and Peek, G.; "Parachute Inflation I – General Phenomenoloy"; lecture delivered at the 2006 H.G. Heinrich Parachute Systems Short Course; May 15 – 19, 2006; copies available on request.

What are the values of C_k?

- **Originally defined as**

$$C_k = \frac{F_{\max}}{\left(\frac{1}{2}\rho V_{stretch}^{\,2}\right)\left(SC_D\right)_{sd}}$$

- **Over the years the values of C_k for different parachute systems have been collected by many authors – see references [1 – 3]**

- **It was found that this C_k data show distinct trends when plotted against the mass ratio R_m**

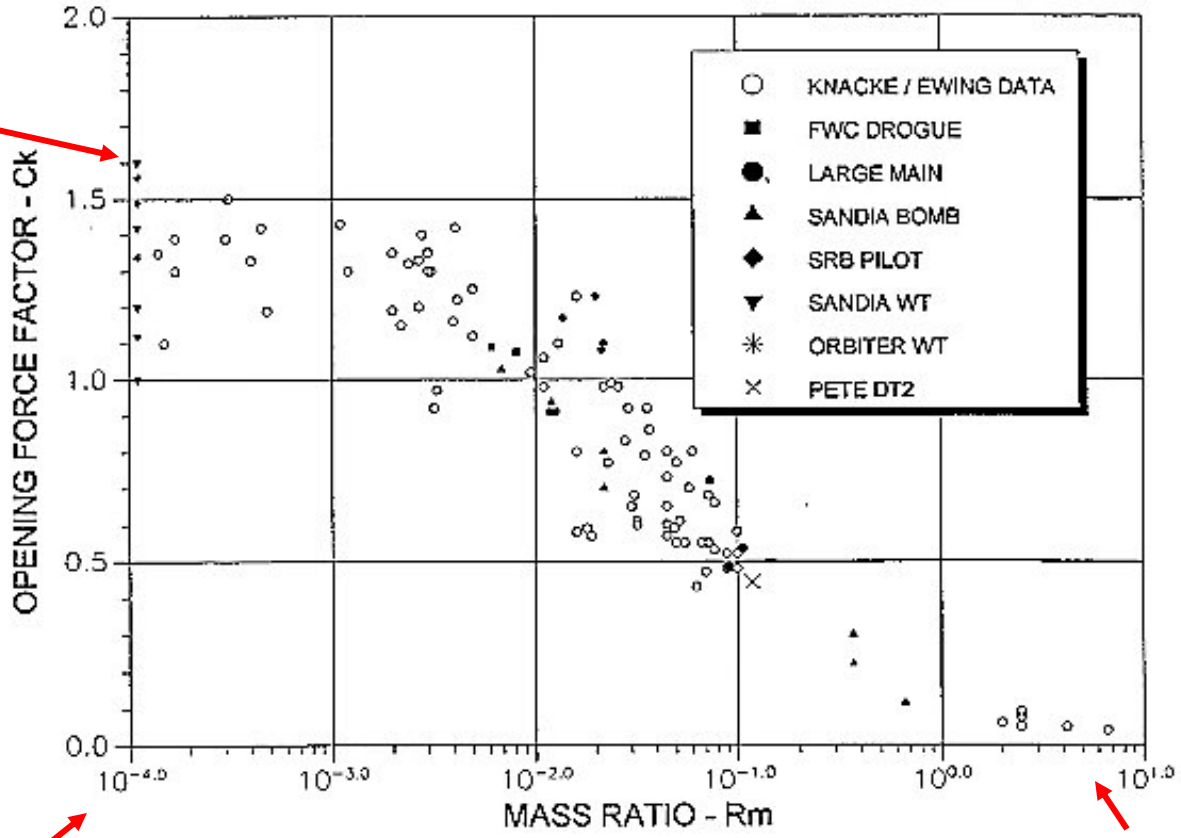
$$R_m = \frac{\rho\left(\left(C_D S\right)_{sd}\right)^{3/2}}{m}$$

Here m = total mass of the payload-parachute system

Opening Shock Factor – Unreefed or permanently reefed hemispherical canopies (low- and high-porosity)

(compiled by Wolf [3])

C_k at $R_m = 0$ is called C_X in Knacke [1]

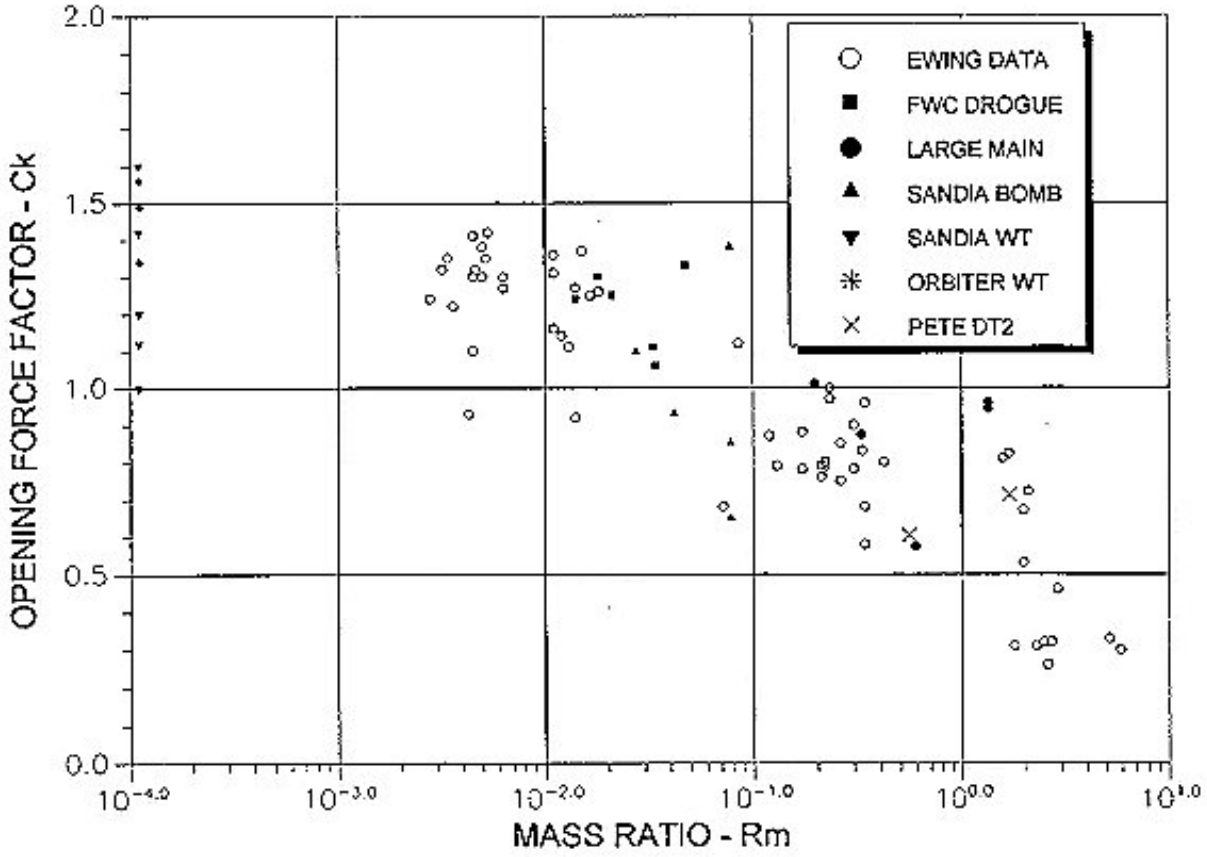


Drogues and pilot chutes;
All chutes deployed in wind tunnels

Personnel and cargo airdrop parachute applications @ low altitudes

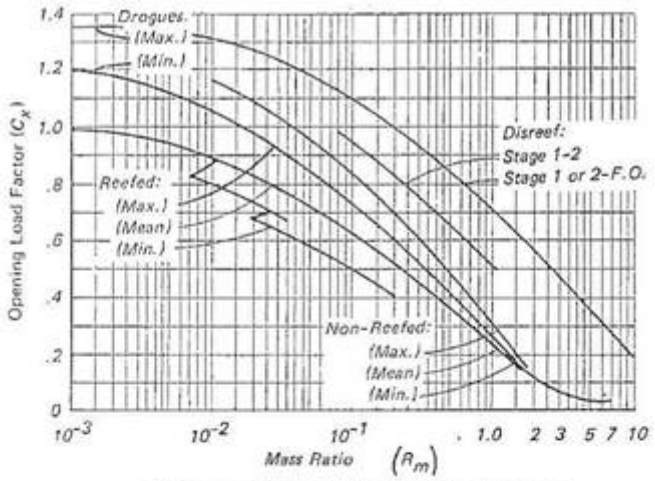
Opening Shock Factor – dis-reefing hemispherical canopies (low- and high-porosity)

(compiled by Wolf[3])

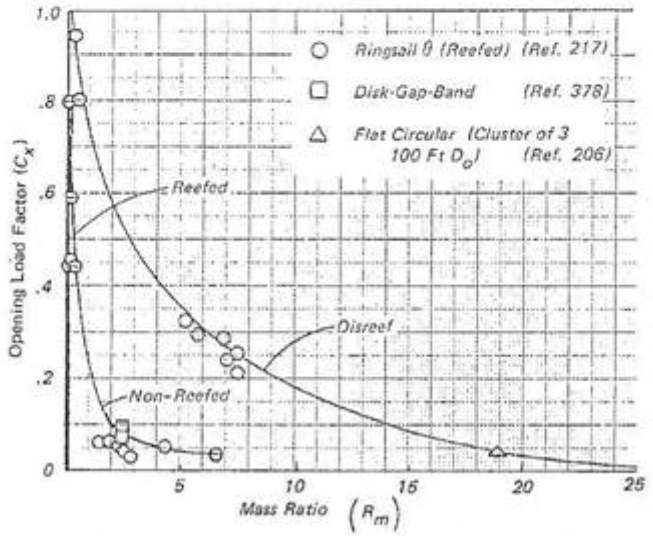


More C_k data

(Figure extracted from
the USAF parachute
design guide – reference [2])



a) Averaged Composite Data From Many Sources



b) Selected Parachute C_x Avg. Disreef to Full and Non-Reefed

Figure 6.25 Parachute Opening Load Factor Vs Mass Ratio

Some important questions about C_k

1) Why is the value of C_k for the "unreefed/permanent reefing" C_k-vs-R_m graph different from that of the "dis-reefing" graph?

2) These graphs were built out of the data collected on hemispherical parachutes; what about the C_k data of parafoils, slider-reefed parachutes and other parachute and reefing types?
Would those data fit in the same two graphs??

Answers

1) Filling time (or "inflation" time) is the key-concept

2) Yes, these two graphs can accommodate the C_k-data of *any* parachute and reefing/dis-reefing type

Back to college physics -

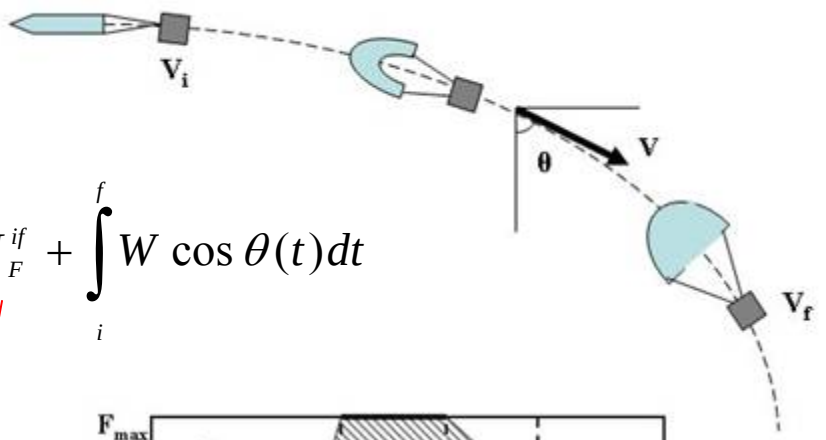
Momentum-Impulse Theorem (details in reference [4])

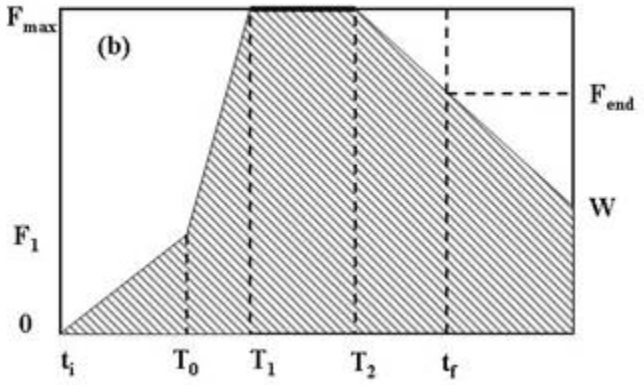
(integral version of F = ma)

$$mV_f - mV_i = impulse = \int_i^f F_D(t)dt + \int_i^f W\cos\theta(t)dt$$

$$\left(mV_f - mV_i\right) = -\underbrace{\left[\frac{1}{2}\rho V_i^2\right](SC_D)_{sd}\,C_k}_{F_{max}} \cdot t_{fill}\, I_F^{if} + \int_i^f W\cos\theta(t)dt$$

$$I_F^{if} = \int_i^f \frac{\left|F_D(t)\right|dt}{F_{max}(t_f - t_i)}$$





I_F^{if} =Drag integral – "area under the curve"

Solve the momentum-impulse equation to get C_k

General Result! Reflects directly F = ma; See ref. [4]

$$C_k = \frac{2}{R_m n_{fill}^{gen}} \Gamma$$

Mass ratio

$$n_{fill}^{gen} \equiv \frac{V_i (t_f - t_i)}{(SC_D)_{sd}^{1/2}} I_F^{if}$$

"Generalized" filling time

$$n_{fill}^{gen} = n_{fill} \frac{D_0}{(SC_D)_{sd}^{1/2}} I_F^{if}$$

$$\Gamma \equiv \frac{-(V_f - V_i) + \int_i^f g \cos\theta(t)\, dt}{V_i}$$

$$= 1 - \frac{V_f}{V_i} \qquad \textit{horizontal}$$

$$= 1 - \frac{V_f}{V_i} + \frac{g}{V_i} t_{fill} \qquad \textit{vertical}$$

Standard filling time

$\sim 0.4 - 0.5$ @ high-R_m -&- $0.2 - 0.4$ @ low-R_m

$$C_k = \frac{2}{R_m n_{fill}^{gen}} \Gamma$$

- **The Momentum-Impulse Theorem confirms the importance of the mass ratio R_m in determining the value of C_k**

- **The Theorem also makes clear that the inflation time (or "filling" time) is another crutial ingredient**

- **OSCALC uses three different concepts of filling time:**

  - **Actual inflation time: t_{fill} (has dimensions of time)**
  - **Standard non-dimensional filling time n_{fill} (has no dimensions)**
  - **Generalized non-dimensional filling time n_{fill}^{gen} (has no dimensions)**

$$n_{fill} = \frac{t_{fill}}{D_0} V_{stretcht} \qquad n_{fill}^{gen} = n_{fill} \frac{D_0}{(SC_D)_{sd}^{1/2}} I_F^{if}$$

Both t_{fill} and D_0 are defined on next slide →

Filling time, or inflation time t_{fill} - *DEFINITION*

- **At high-R_m: t_{fill} = (dimensional) time interval elapsed between line stretch and the moment when the canopy reaches its designed steady-state diameter for the first time (as the parachute often-times over-inflates)**

- **At low-R_m: t_{fill} = (dimensional) time interval elapsed between line-stretch and the moment when the canopy reaches its maximum extension**

- **D_0 = "nominal" canopy diameter (hemisphericals) = $(4S_0/\pi)^{1/2}$**
  with S_0 = total canopy surface area, inlcuding the area of the vents and all other openings [1]

- **D_0 = diameter of circle of same area as wing *span* x *chord* (parafoils)**

- **There is a lot of data on the non-dimensional filling time n_{fill} – for details see Section 4 below**

- **Note that t_{fill} can be obtained from the analysis of the video of the inflation process; n_{fill} is then calculated, using this value of t_{fill}, as well as the value of D_0 and an estimate of $V_{stretch}$**

How is inflation time used in the C_k-R_m graphs?

- **One chooses the "right" graph according to the value of the generalized filling time n_{fill}^{gen} - see next two slides (the "colored" data points were collected by GP and JP††). More details can be found in reference [4].**

→ **If $n_{fill}^{gen} \geq 4$, the user chooses the graph formely associated with *un-reefed and reefed* in Wolf's compilation**

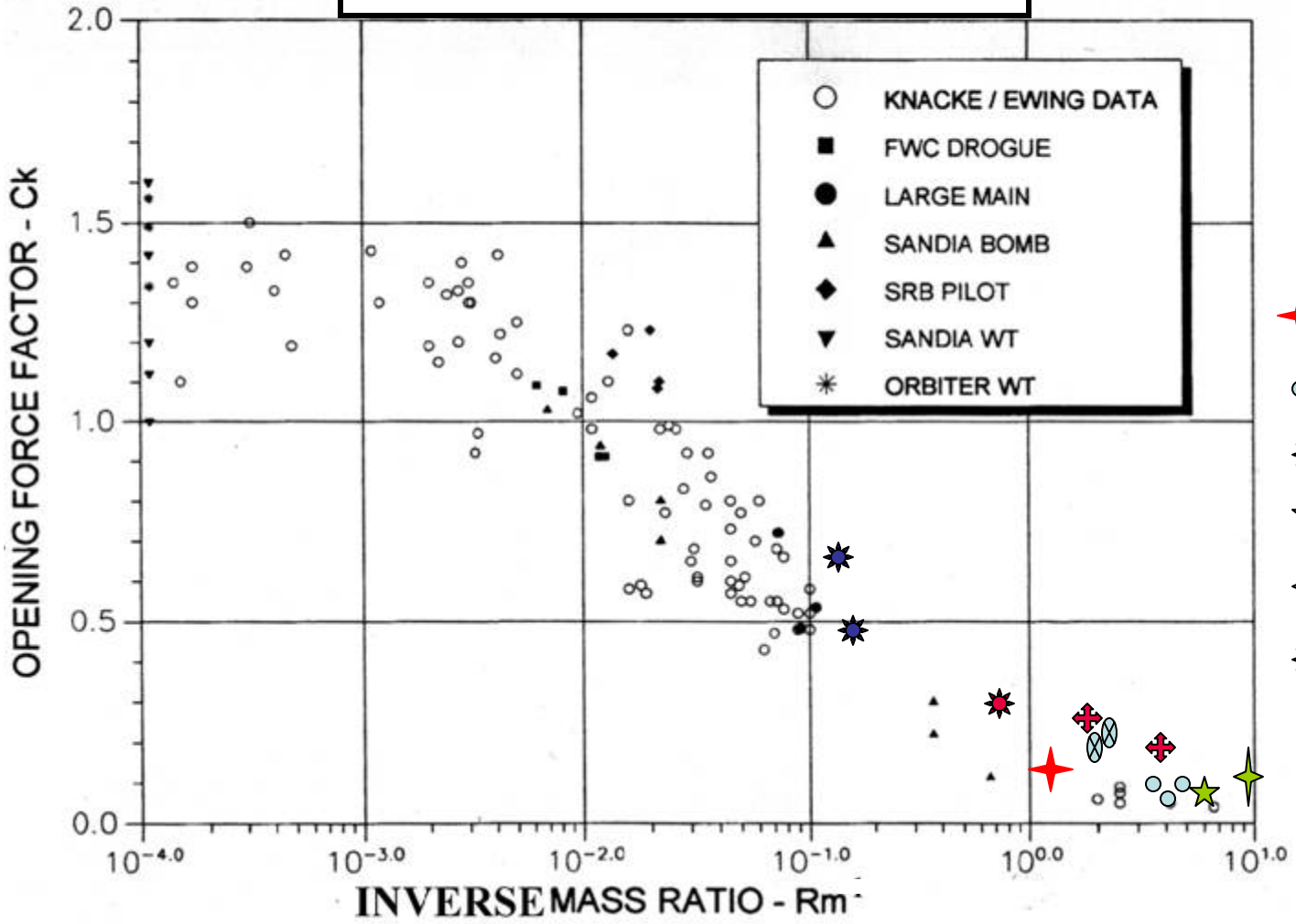
→ **If $1 \leq n_{fill}^{gen} < 4$, the user chooses the graph formely associated with *dis-reefing* in Wolf's compilation**

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††Note: The "half-scale C-9 - unreefed" data shown on the small-n_{fill} graph characterized a parachute system with unusually large riser separation, which yielded unusually shorter filling times

Original graph from: D. Wolf - paper AIAA-99-1702; "colored" data points collected by GP and JP

$$n_{fill}^{gen} \geq 4$$



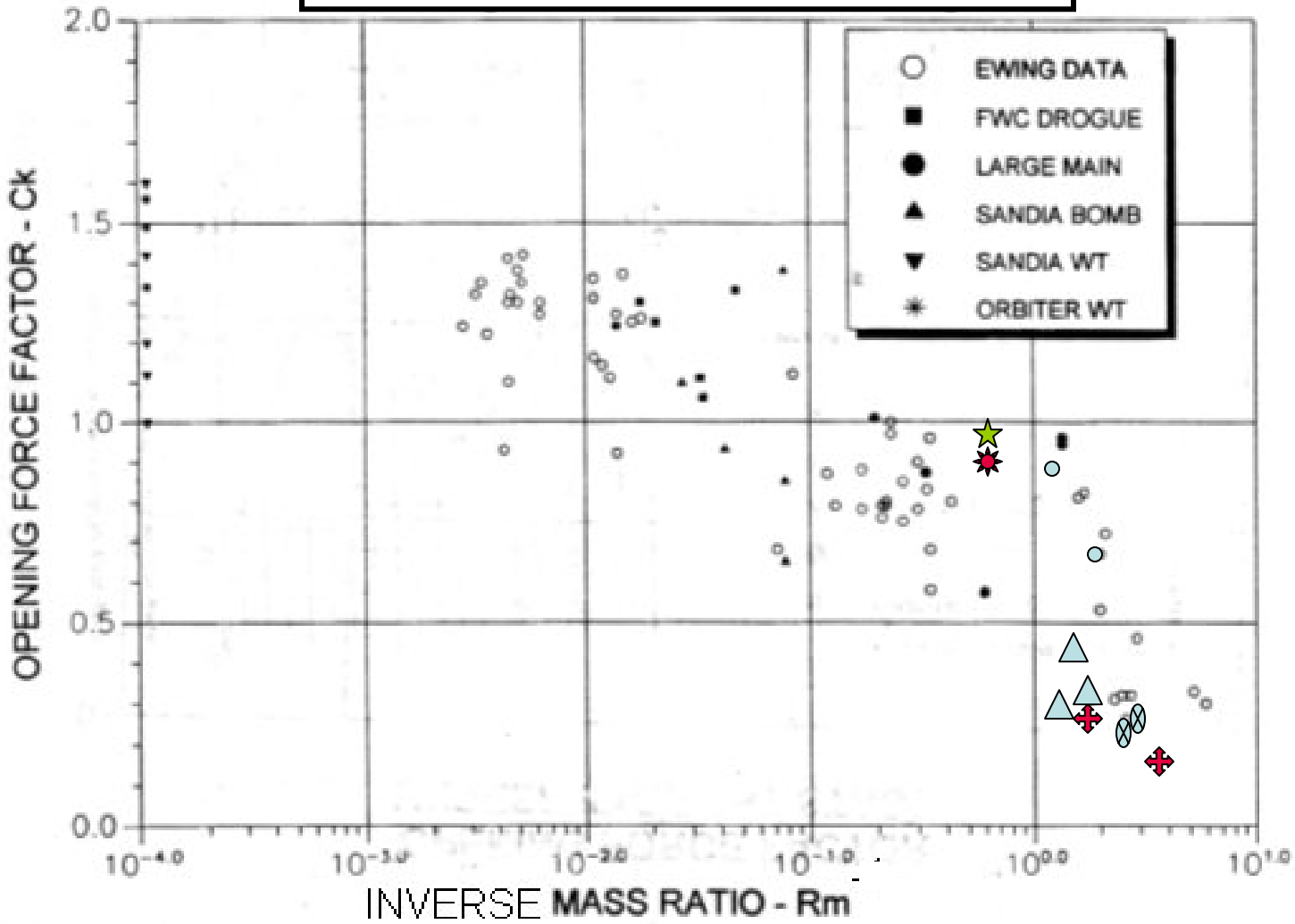
- Full-scale TCD (Deep cone)
- Unreefed C-9
- Reefed C-9 (24%)
- T-10C
- MC1-1C
- Half-scale C-9; Reefed at 16%

Aeroconical/steerable – National Phantom

Aeroconical/steerable – GQ Security Crossbow

Original graph from: D. Wolf - paper AIAA-99-1702; "colored" data points collected by GP and JP

$$1 \leq n_{fill}^{gen} < 4$$



Aeroconical/steerable – GQ Security Crossbow

Aeroconical/steerable – Pioneer K22

Aeroconical/steerable – National Phantom

Half-scale C-9 - unreefed

Parafoil - ParaFlite Strato Cloud – no slider

Parafoil - Precision Aero Falcon 175 – no slider

- **NOTE: OSCALC uses the following approximation for calculating the value of I_F^{if}**
  - **If $R_m > 0.10$ then $I_F^{if} = 0.5$**
  - **If $R_m < 0.01$ then $I_F^{if} = 0.2$**
  - **If $0.01 \leq R_m \leq 0.10$ then $I_F^{if} = (0.5 + 0.2)/2$**
- **NOTE: The integral Γ is one of the factors that generates the scatter in the two graphs; OSCALC does not computes Γ but accounts for it through the C_k-lower and upper bounds that must be entered**

$$C_k = \frac{2}{R_m n_{fill}^{gen}} \Gamma$$

$$n_{fill}^{gen,\,acft} = n_{fill} \frac{D_0}{(SC_D)_{sd}^{1/2}} I_F^{if}$$

To recap

- The value of n_{fill} determine which one of the two C_k vs. R_m graphs to use

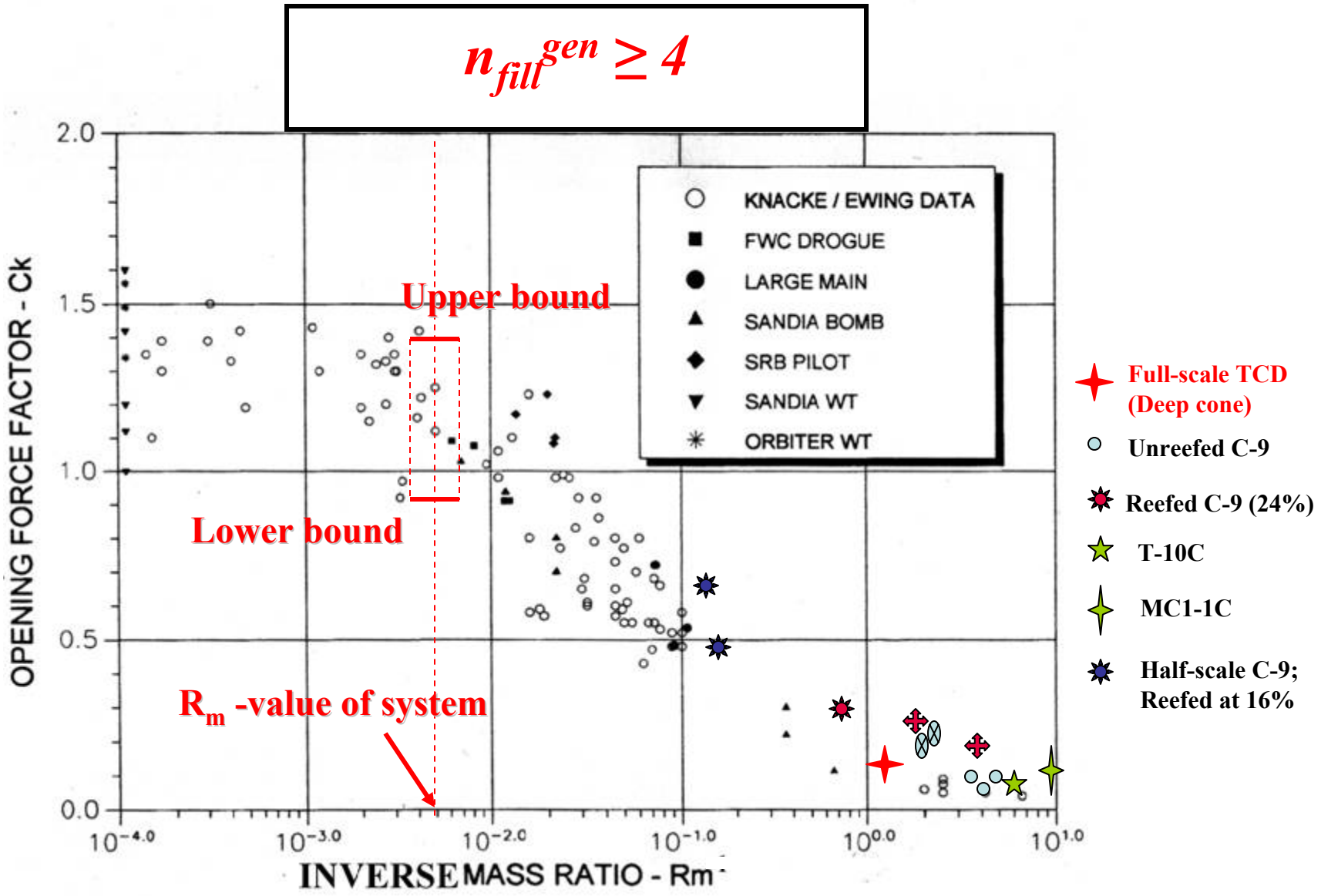
- From the chosen graph, the value of C_k, of its lower bound and of its upper bound are obtained by spotting the R_m–value of the system under consideration - see next slide for example

- F_{max} is finally calculated from this C_k data and from the other basic inputs $(SC_D)_{sd}$, ρ and $V_{stretch}$; its upper and lower bounds are estimated as well, using the upper and lower bounds of C_k

- Note: Section 7 explains what to do when $n_{fill}^{gen} < 1$

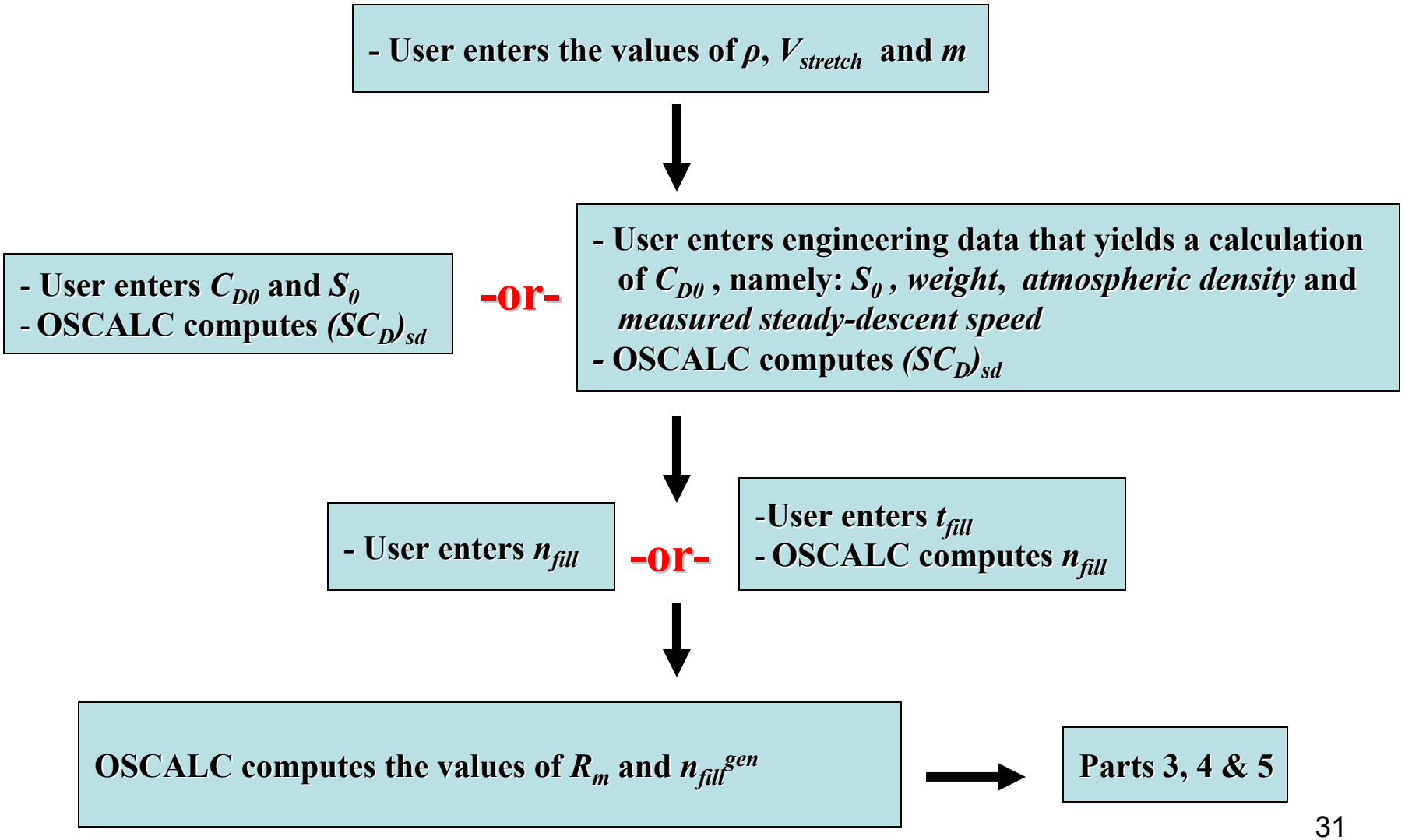
$$F_{\max} = \left(\frac{1}{2}\rho V_{stretch}^{\ 2}\right)\left(SC_D\right)_{sd} C_k$$

Original graph from: D. Wolf - paper AIAA-99-1702; "colored" data points collected by GP and JP



3. How to run OSCALC?

3. How to run OSCALC? The logic flow is as follows: Parts 1 & 2 – Input & intermediate calculation



Parts 3, 4 & 5. Choice of the proper $C_k - R_m$ graph; input of C_k; final calculation

Parts 1 & 2

↓

- User chooses either the high- or low-n_{fill}^{gen} graph of C_k vs. R_m, based on the computed value of n_{fill}^{gen}
- From the graph the user picks the "average" value of C_k , as well as its upper and lower bounds
- User enters those three values of C_k into OSCALC

↓

OSCALC computes F_{max} and its upper- and lower-bounds

How to run OSCALC?†

- Copy the file named OSCALC.exe in a directory of your choice; OSCALC does not require any input files, nor does it generate any ouput files

- Run OSCALC.exe

- The main window appears with one graph tucked under
  – see figure 3.1 below

† These instructions apply both to V1.0 and V1.01

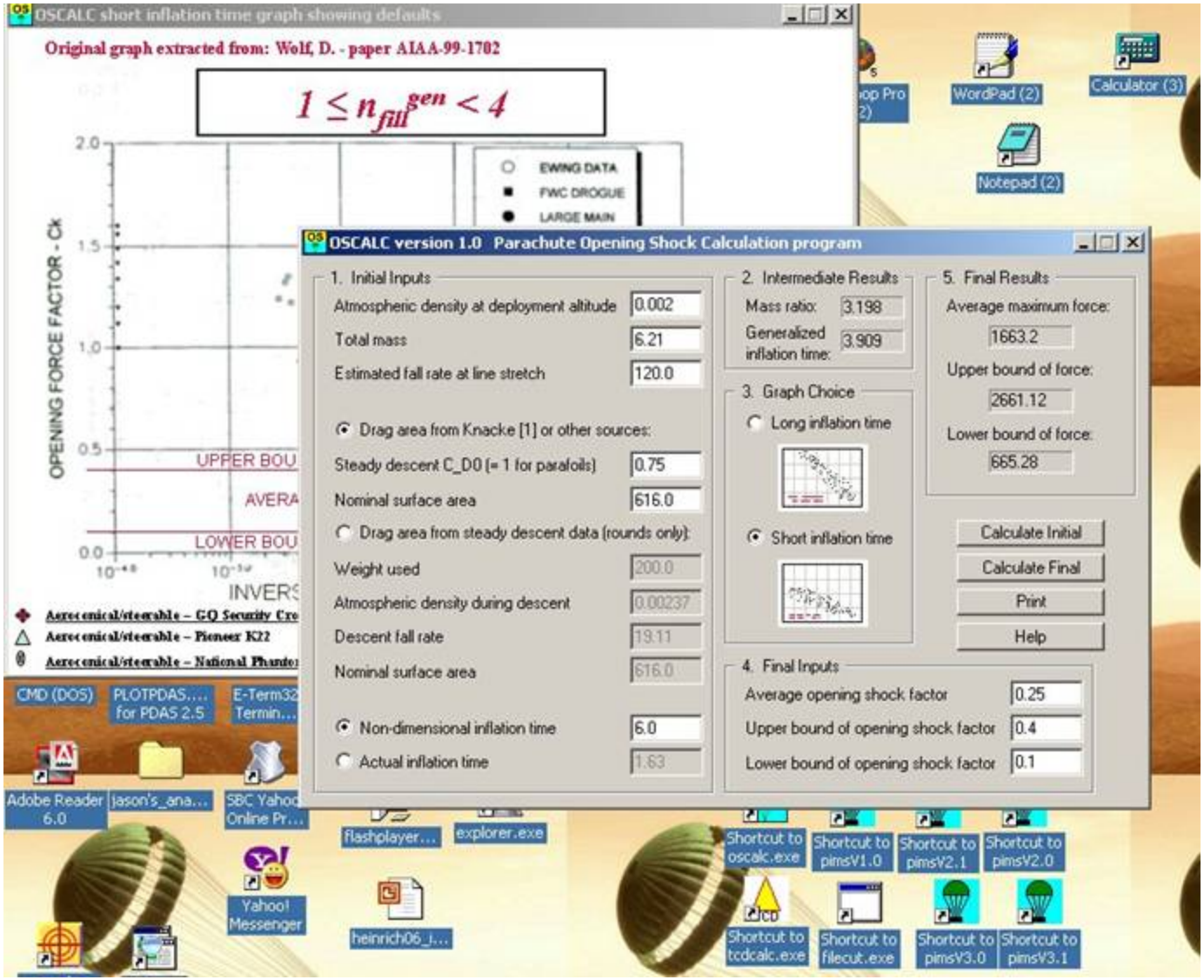


Figure 3.1 "Default" calculation and windows displayed at the beginning an OSCALC session

How to run OSCALC? Cont'd

- This window is the "Default" window; the graph shows how the values of C_k were picked, given the mass ratio and inflation time that resulted from the default input values. The Default case is discussed further in Section 6 (example 6.1)

- Click on the graph to see it completely; click on the program window to return to the program

- The correspondence between the input/output shown on the main window and the variables discussed in this manual is shown in figures 3.2, 3.3 and 3.4 below

- Note about the Default graph – the graph will be replaced by the same graph, but without comments, whenever the "Short Inflation Time" radio-button is clicked. Exit OSCALC and start a new session if there is a need to look at the original Default graph again

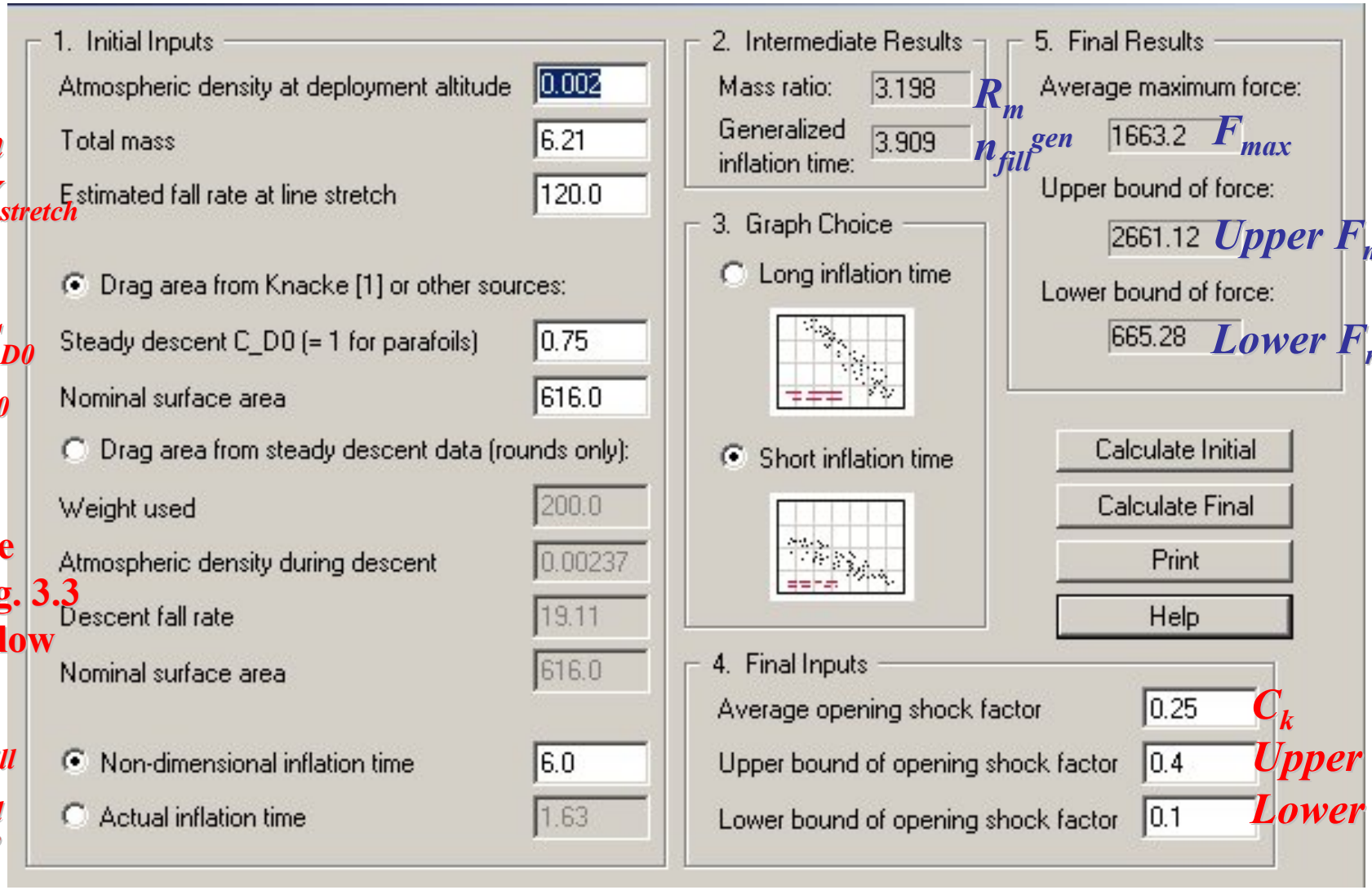


Figure 3.2 Basic input and output of OSCALC. The color coding for this figure is as follows: Calculated versus input.

- **Again, note that the user can either:**

- **enter the values of C_{D0} and S_0 and have OSCALC compute the value of $(SC_D)_{sd} = C_{D0}\, S_0$ (in the case of parafoils, enter $C_{D0} = 1$ & $S_0 = chord * span$)**

- **enter proprietary engineering data related to the steady descent of the canopy in order for OSCALC to compute C_{D0} and $(SC_D)_{sd} = C_{D0}\, S_0$ (note: W_{sd} and ρ_{sd} *can be* different from the values of $m(g)$ and ρ used for the calculation of F_{max}). Note: this alternative applies only to hemispherical parachutes.**

Just click the relevant radio-button (only one button can be clicked during the same run)

$$C_{D0} = \frac{2W_{sd}}{\rho_{sd} S_0 V_{sd}^{\,2}}$$

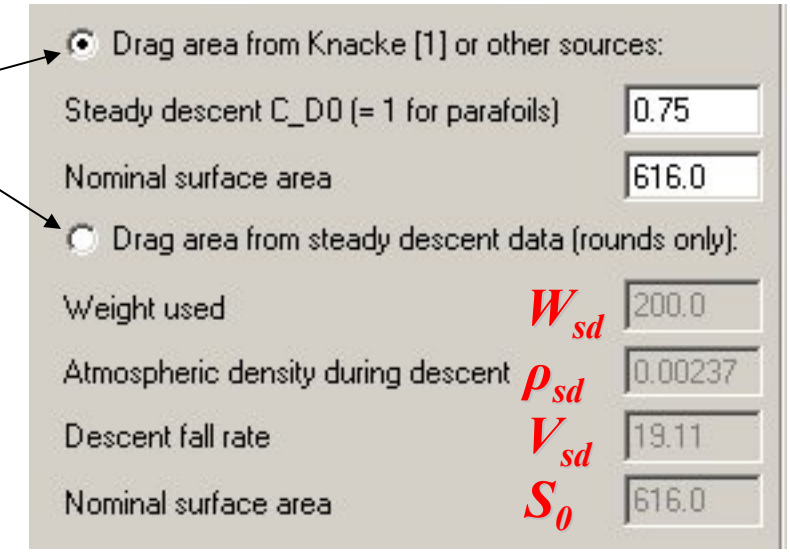


Figure 3.3 Drag area options

• **Note also that the user can either:**

- enter the values of n_{fill}

or…

- enter the dimensional inflation time t_{fill} (as measured on video); OSCALC will then compute n_{fill}

Just click the relevant radio-button (only one button can be clicked during the same run)

$$n_{fill} = \frac{t_{fill}}{D_0} V_{stretcht}$$

$$D_0 = \sqrt{4S_0 / \pi}$$

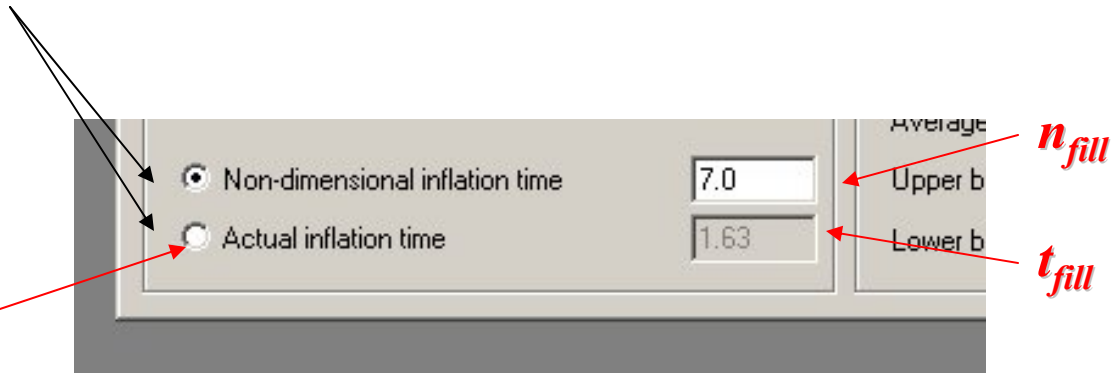


Figure 3.4 Inflation time options

- After entering the input data in "1. Initial inputs", click the "Calculate Initial" button

- After looking at the values of R_m and n_{fill}^{gen} in "2. Intermediate results", click the relevant graph in "3. Graph choice" in order to find C_k and its bounds; enter these values in "4. Final inputs". Then click the "Calculate Final" button and look at the calculated force in "5. Final results"

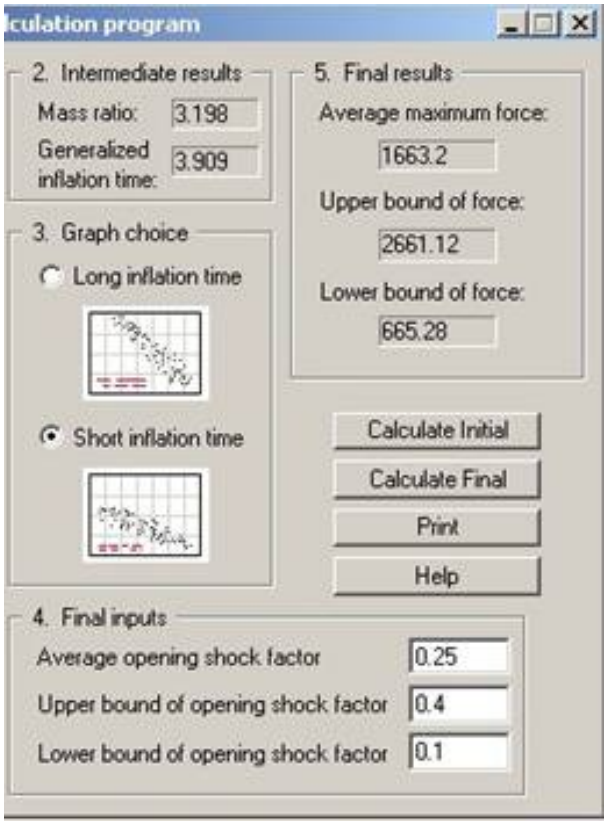


Figure 3.5 Parts 3, 4 and 5 of an OSCALC run

- **Note: the button "Print" sends to the printer an image of the worksheet (but not of the graph)**

- **Note: the button "Help" generates a window that shows copyright information, a list of suggested units, an acknowledgement and limited information on the program**

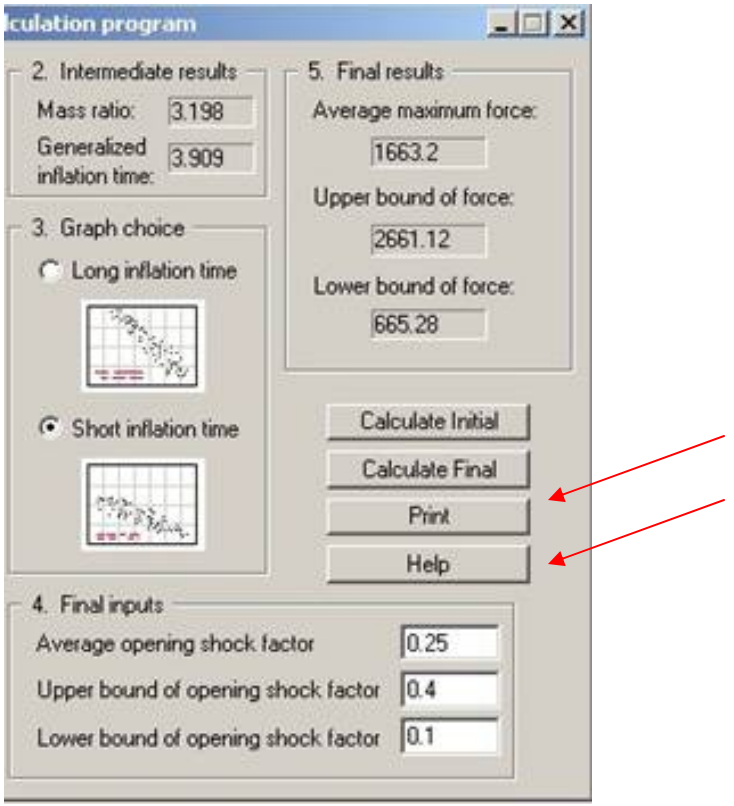


Figure 3.6 The "Print" and "Help"buttons

4. More information on filling time

- **The following references contain experimental data on the inflation time of several types of parachutes.**

- **References [1] and [2] previously mentioned also contain inflation time data**

4. References on inflation time

[6] Lee, C. K., "Modeling of Parachute Opening: an Experimental Investigation", *Journal of Aircraft*, 26, 444 – 451, 1989.
[7] Cruz, J. R., Kandis, M. and Witkowski, A.; "Opening Loads Analyses for Various Disk-Gap-Band Parachutes"; paper AIAA 2003-2131. *17thAIAA Aerodynamic Decelerator Systems Technology Conference and Seminar*, Monterey, CA, May 19 -22, 2003.
[8] Johari, H., and Desabrais, K. J.; "Stiffness Scaling for Solid-Cloth Parachutes"; *Journal of Aircraft*, 40, pp. 631 – 637, 2003.
[9] Berndt R. J. and DeWeese J. H., "A Filling Time Prediction Approach for Solid Cloth Type Parachute Canopies", 2nd AIAA Aerodynamic Decelerator Systems Technology Conference, Houston, TX, September 7-9, 1966, pp. 17-32.
[10] Wolf, D.; "A Simplified Dynamic Model of Parachute Inflation"; *Journal of Aircraft*, 11, No. 1, pp. 28 - 33, 1974.

4. References on inflation time (cont'd)

[11] Lingard, J. S.; "A Semi-empirical Theory to Predict the Load-time History of an Inflating Parachute"; AIAA-84-0814, 8th AIAA Aerodynamic Decelerator and Balloon Technology Conference, 1984, Hyannis, MA, April 2-4, 1984.
[12] Lee, C. K.; "Experimental Investigation of Full-Scale and Model Parachute Opening"; paper AIAA 84-0820; 8th AIAA Aerodynamic Decelerator and Balloon Technology Conference, 1984, Hyannis, MA, April 2-4, 1984.
[13] Lee, C. K., Lanza, J. and Buckley, J.; "Experimental Investigation of Clustered Parachute Inflation"; paper AIAA 97-1478. *14thAIAA Aerodynamic Decelerator Systems Technology Conference and Seminar*, San Francisco, CA, May 19 -22, 1997.
[14] Lingard, S. J.; *Ram-Air Parachute Design*; AIAA Aerodynamic Decelerator Systems Technology Seminar; May, 1995; 63 pp. Unpublished.

4. References on inflation time (cont'd)

[15] Barnard, G. A.; "The Effect of Extreme Altitude of Parachute Filling Distance"; AIAA-93-1207; 12th AIAA Aerodynamic Decelerator Systems Technology Conference and Seminar (RAeS/AIAA) London, England, May 10-13, 1993.
[16] Heinrich, H.G.; "The Opening Time of Parachutes Under Infinite Mass Conditions"; AIAA-68-12; 3rd AIAA Aerodynamic Decelerator Systems Technology Conference, El Centro, CA, September 23-25, 1968.
[17] Heinrich, H.G.; "The Opening Time of Parachutes Under Infinite Mass Conditions"; *Journal of Aircraft*; **6**, No.3, pp. 268 – 272; 1969.
[18] J. Potvin and G. E. Peek; "Inflation and Steady-Descent Characteristics of Truncated Cone Decelerators"; AIAA paper 2005-1620; 18th AIAA Aerodynamic Decelerator Systems Conference and Seminar, Munich, Germany, May 23-26, 2005.

- **The following tables summarize some of the data found in these references. Please go to the references to get the details on the parachute construction dimensions, payload characteristics and drop conditions**

Low- and high porosity hemispherical - see Knacke [1]

Various Parachute Types.

Parachute type	Canopy fill constant, n		
	Reefed opening	Disreef opening	Unreefed opening
Solid flat circular	ID[a]	ID	8
Extended-skirt, 10%	16-18	4-5	10
Extended-skirt, full	16-18	7	12
Cross	ID	ID	11.7
Ribbon	10	6	14
Ringslot	ID	ID	14
Ringsail	7-8	2	7
Ribless guide surface	. . .	. . .	4-6

[a] ID = Insufficient data available for meaningful evaluation.

Note about the previous Table –

Why the differences in n_{fill} between reefed opening, disreef opening and unreefed opening?

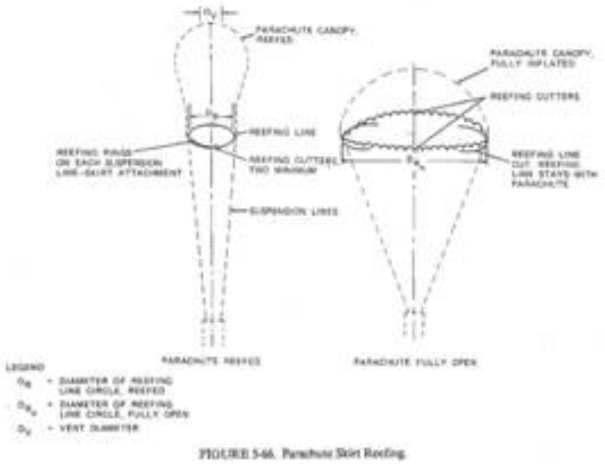


FIGURE 5-66. Parachute Skirt Reefing

Relative to unreefed opening….

- **reefed opening involves a canopy mouth which over time does not open as widely – n_{fill} should be larger (longer inflation time); However! n_{fill} can be shorter as well, with parachutes reefed by very short reefing lines: small reefing ratios result in inflated canopy volumes that are *smaller* than the inflated volumes of the unreefed configurations.**

- **disreefed opening needs a smaller amount of air to fill the canopy from its initial state (i.e. inflated but "reefed") to it final state (i.e. fully opened) – n_{fill} should be smaller (shorter inflation time)**

Deep cone

Size	Total weight, deploy altitude, V_{init}	Reefing	n_{fill}	References
Deep Cone Cone height = 16.00ft Cone base = 10.66ft D_0 = 32.72ft (hypothenuse of half cone)	18lbs; 700ft MSL; 110ft/sec	No reefing	6 - 10	See reference [18];calculation based on D_0



Parafoils

Size	Total weight, deploy altitude, V_{init}	Reefing	n_{fill}	References
Parafoil Paraflite Strato Cloud chord = 10.08ft span = 19.16	160lbs; 700ft MSL; 105ft/sec	None	2.8	
Parafoil PD Stiletto150 (skydiving/elliptical) 1990's design 9 cells – 150ft^2 factory configuration	200lbs; 4000ft MSL; 160 - 170ft/sec	Slider (sail/free)	16 – 20 (Typical drop-to-drop variations)	
Parafoil PD Sabre150 (skydiving/elliptical) 1990's design 9 cells – 150ft^2 factory configuration	200lbs; 4000ft MSL; 160 - 170ft/sec	Slider (sail/free)	10 -19 (Typical drop-to-drop variations)	

Note: with slider-reefed systems, n_{fill} will depend on the actual dimensions of canopy rigging angle; brake setting; inlet, canopy and slider design and dimensions

Parafoils

Size	Total weight, deploy altitude, V_{init}	Reefing	n_{fill}	References
Parafoil Paraflite MT-1X (skydiving/rectangular) 1980's design 7 cells – 370ft^2 factory configuration	274-360lbs; 10,000ft MSL; 270-310ft/sec	Slider (both sail/free and pilot-chute-controlled)	12 -19 (Typical drop-to-drop variations)	
Parafoil No data; 1980's design for personnel applications	No data	Slider (sail/free)	14	Reference [14]
Parafoil No data; 1980's designfor personnel applications	No data	No reefing	3.5	Reference [14]

Note: on slider-reefed systems, suspension line-abrasion can significantly increase the value of n_{fill}, to values well-exceeding $n_{fill} = 20$

Remarks

- These tables show that n_{fill} depends on canopy design and reefing

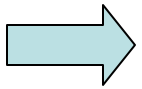
- BUT n_{fill} also depends on how wide the canopy mouth is opened at the beginning of the inflation process (time of line stretch in many cases)

- wide mouth = lots of air gulped-in fast → fast inflation – low value of n_{fill}

- narrow mouth = air entering at a slower rate → slow inflation – large value of n_{fill}

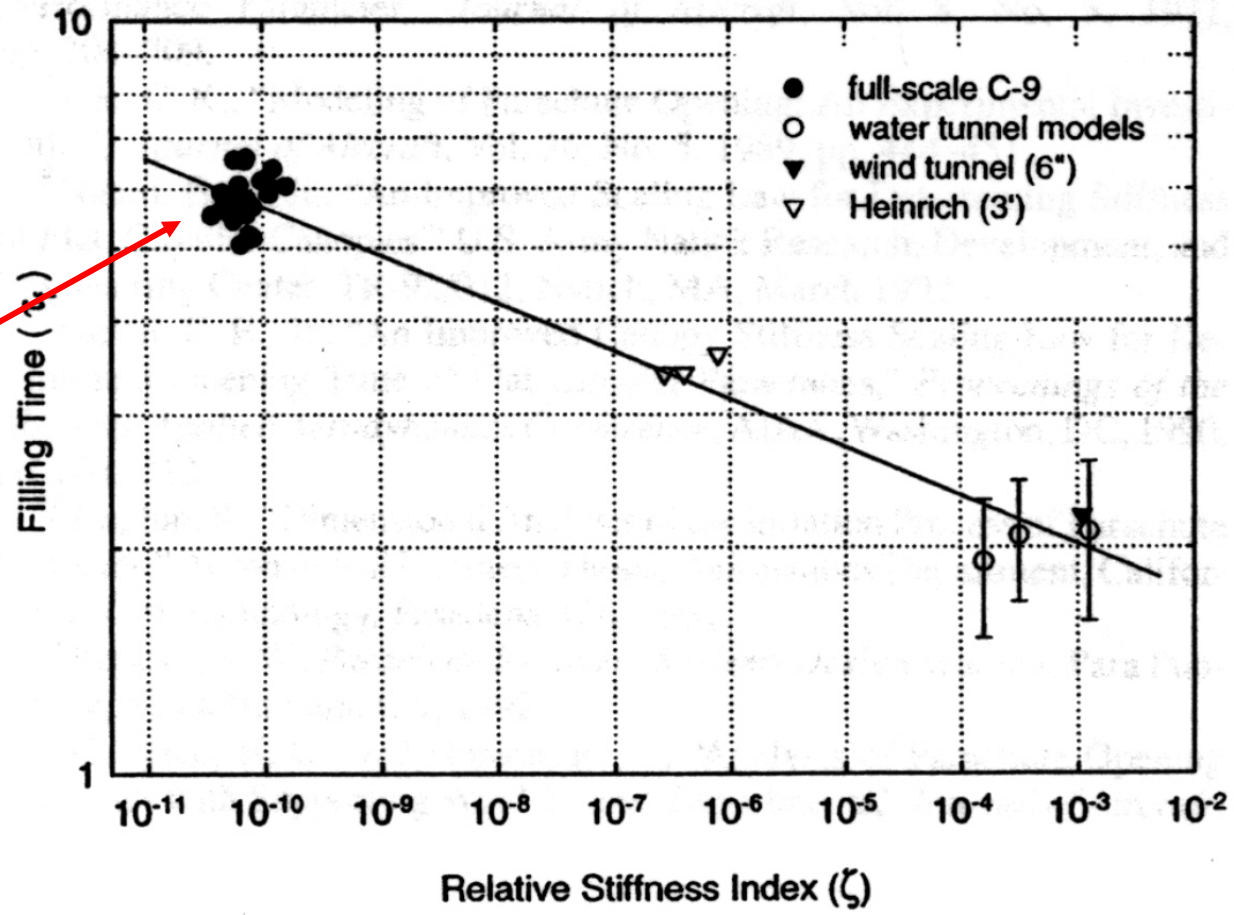
- The actual amount of opened mouth area is often and strongly determined by the way the parachute gets out of its container bag and aligns into the wind – n_{fill} depends on what happened during <u>deployment!</u>

- Expect to see drop-to-drop variations of n_{fill} *– next slide*



Drop-to-drop variations during test drops, from $n_{fill} \sim 5$ to $n_{fill} \sim 7$

(Figure by Desabrais & Johari [8])



More remarks

- **n_{fill} depends on how wide the canopy mouth is opened at the beginning of the inflation process –oftentimes a random process due to the canopy being in a limp state (i.e. non-tension bearing) as it is exposed to the wind**

- **Sometimes such mouth state is determined by design (payload width, reefing, etc.). Check out the following formula, which follows from this definition of the filling time:**

Fill time ~ Needed air to fill canopy volume / (air speed x mouth opened area)

…which together with the definition of n_{fill} implies:

$$n_{fill} = \frac{\forall_{tofill}}{S_{init}^{netmouth} D_0}$$

$$n_{fill} = \frac{\forall_{tofill}}{S_{init}^{netmouth} D_0}$$

- **Note: In most cases the "to-fill" volume V_{tofill} is impossible to calculate a priori because of porosity (no one can track accurately the air that goes in and then later leaves)**

- **The formula is more useful in studies where the scaling properties of V_{tofill} are known, eventhough its precise value is not; see examples on next the slide**

• **Here how the mouth state is determined by design**

$$n_{fill} = \frac{\forall_{tofill}}{S_{init}^{netmouth} D_0}$$

Example 1: comparison of the filling time between a full-scale and a half-scale canopy, hooked to the *same* (i.e. "unscaled") payload container

$$n_{fill}^{half\ scale} \sim (1/4)\ n_{fill}^{full\ scale}$$

Example 2: comparison of the filling time of a given *low-porosity* canopy involved in:

→ unreefed inflation

versus

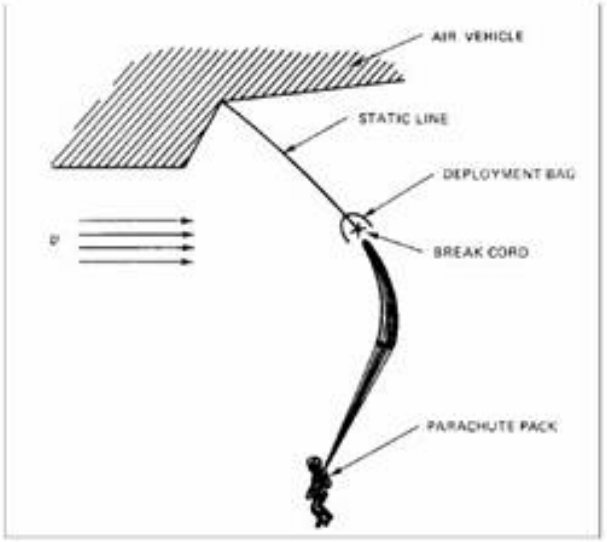
→ dis-reef inflation, from a very small reefing ratio (~20%)

$$\frac{n_{fill}^{disreef}}{n_{fill}^{unreef}} \sim 29 \frac{S_{init}^{mouth-unreef}}{D_0^2} \sim 0.29$$

Filling time and deployment strategy

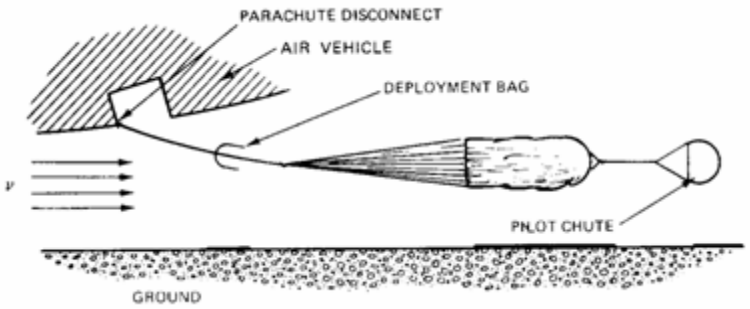
- **n_{fill} will vary depending on whether the parachute is deployed in a cross-wind or facing the wind. Typically n_{fill} will be larger, *and with more drop-to-drop variations*, with the former approach than with the latter. See next slide.**

- **In cross-wind deployments the canopy mouth is kept shut for a while longer during the early stages of inflation**



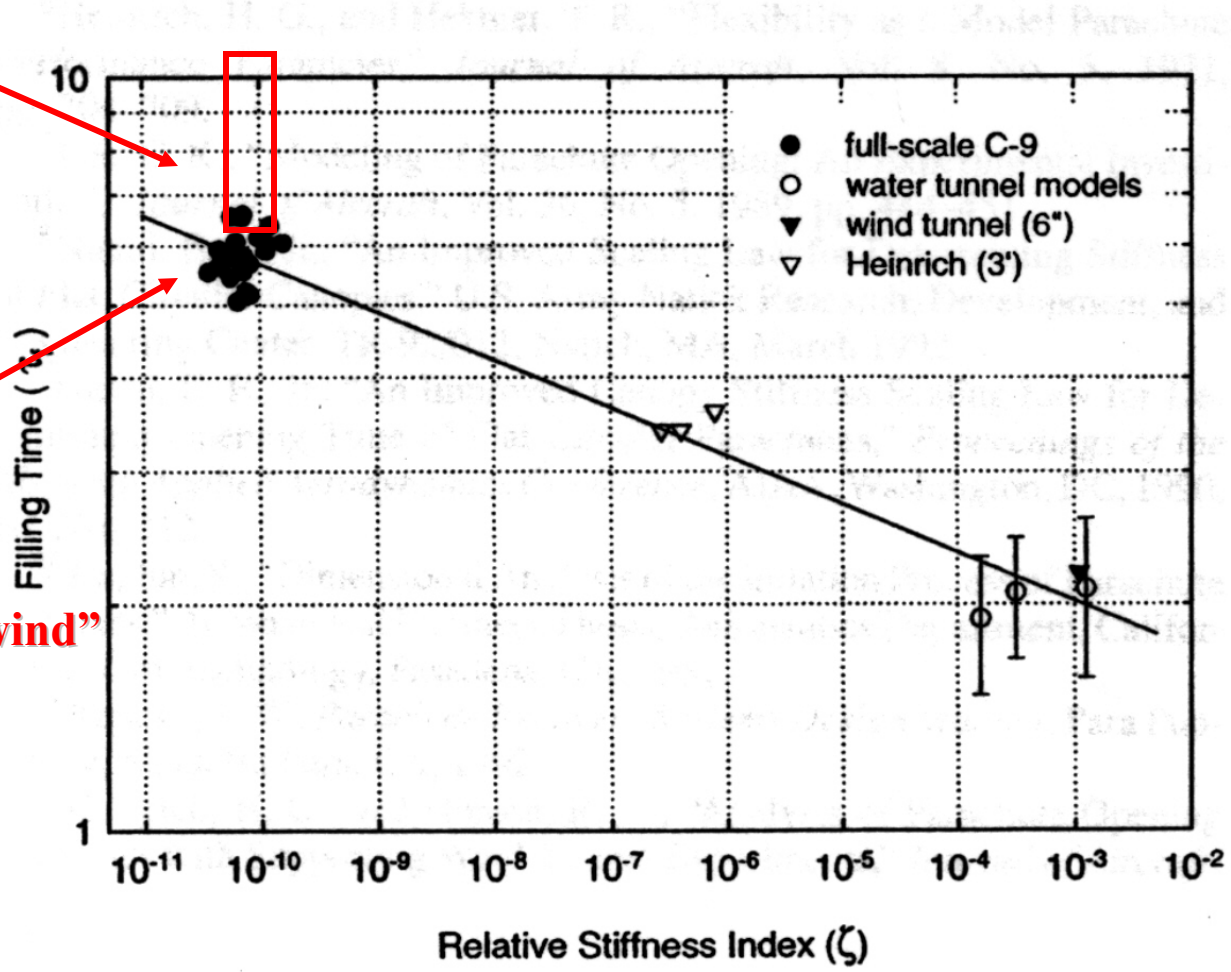
Cross-wind

Facing the wind



C-9 deployments: Data by GP and JP; "cross-wind"

Drop-to-drop variations during test drops, from $n_{fill} \sim 5$ to $n_{fill} \sim 7$ – "facing the wind" deployments



(Figure by Desabrais & Johari [8])

Other dependences

• **Dependence on mass ratio – example [11]**

$$n_{fill} = R_m 0.0501 + 5.68$$

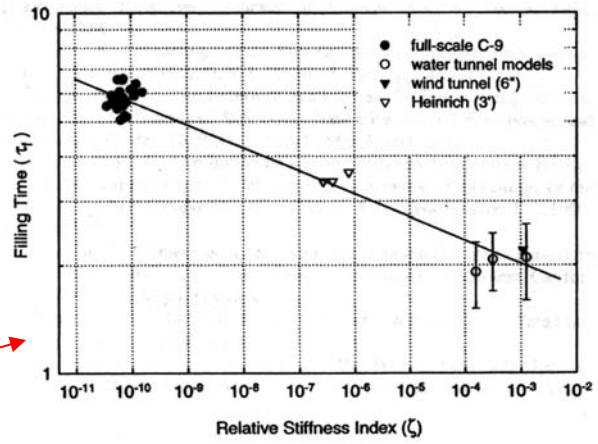
USAF C-9 (low-porosity flat-circular)

Yes, that makes sense: n_{fill} should be smaller if the parachute-payload moves at constant speed, compared to the same system undergoing a deceleration – in other words the air in-flux is larger with the former than with the latter.
In the small R_m- regime the dependence may not be as dramatic.

• **Dependence on relative stiffness: important issue when working with small & sub-scale models**

$$n_{fill} = 1.3\varsigma^{-0.064} = 1.3 \left[\frac{E}{\rho V_{stretch}^2 (1-\nu^2)} \frac{\delta}{D_0} \right]^{-0.064}$$

Tunnel data by Heinrich and also
By Johari and Desabrais [8]



5. Some tricks to estimate the fall speed at line stretch

5. Some tricks to estimate the fall speed at line stretch

- $V_{stretch}$ = parachute-payload fall rate @ the beginning of inflation
- Most often ≠ $V_{aircraft}$
- Depends on deployment strategy:

$$V_{stretch} \approx \sqrt{\frac{2W}{\rho\left(S_{drogue}\right)C_D^{drogue}}}$$

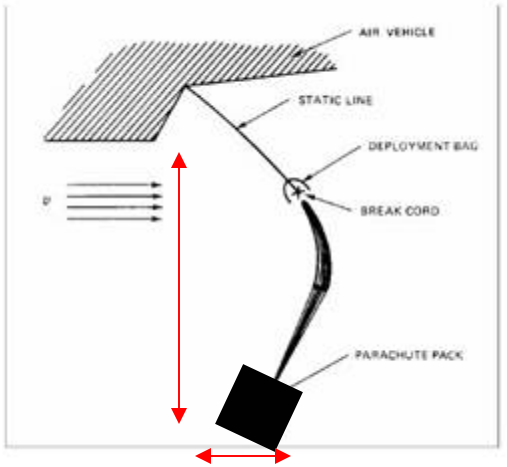
  - After long, stabilized freefall

  - Deployment by static line and for a mostly pre-inflation horizontal trajectory

$$V_{stretch} = 2\frac{\Delta X}{\Delta t} - V_{acft}$$

1-dim; const-acceleration kinematics:
ΔX = acft-payload distance
Δt = deployment duration

- **Deployment static line and for a mostly-vertical pre-inflation trajectory**



Adjust "FUDGE" to match mean flight angle seen on video – payload drag is slowing-down horizontal motion of payload over time

$$V_{stretch} = \sqrt{\left(V_{stretch_horiz}\right)^2 + \left(V_{stretch_vertical}\right)^2}$$

$$\left(V_{stretch_horiz}\right) \sim (FUDGE) \cdot V_{acft}$$

$$\left(V_{stretch_vertical}\right) \sim \sqrt{2g\left(L_{staticline} + \frac{D_0}{2} + L_{susplines}\right)}$$

Vertical acceleration dominated by gravity

6. Examples

6. Examples

6.1 Unreefed low-porosity hemispherical canopy ("Default" case)

6.2 Permanently reefed high-porosity hemispherical canopy, deployed at high altitude and at low mass ratio

6.3 Un-reefed parafoil vs. dis-reefing parafoil

6.4 Dis-reefing hemispherical canopy

6.5 Parachute cluster (unreefed)

6.6 What to do with a new design that is not documented in the World's database on inflation time and opening shock factor

6.1 Unreefed low-porosity hemispherical canopy ("Default" case)

This example would be typical of the USAF C-9 canopy - a "flat circular" canopy made of low permeability fabric. This canopy has no drive slots – just a small vent at the apex.

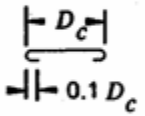
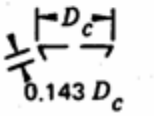
ρ = 0.002 sl/ft^3 – deployment at 5000ft MSL
m = 6.21 sl (corresponding to 200 lbs on Earth)
$V_{stretch}$ = 120 ft/sec
C_{D0} = *0.75* from reference [1] - see figure 6.1
S_0 = *616 ft^2* – calculated from $S_0 = \pi\ (28ft)^2/4$

n_{fill} = 6 – See figure 6.2

OSCALC gives: R_m = 3.19 and n_{fill}^{gen} = 3.91
User should choose the "Short Inflation Time" graph
User would pick C_k = 0.25 with ~ 0.4 and ~ 0.1 as bounds
See figures 3.1 and 6.3

The result is F_{max} = 1663.2 lbs with 2661.1 lbs and 665.3 lbs as bounds

Figure 6.1. Where to get steady-state drag coefficient data?
Go to Knacke [1] Tables 5-1 – 5-4.
Figure below is from Table 5-1

TYPE	CONSTRUCTED SHAPE PLAN	CONSTRUCTED SHAPE PROFILE	$\frac{D_c}{D_o}$	INFLATED SHAPE $\frac{D_p}{D_o}$	DRAG COEF C_{D_o} RANGE	OPENING FORCE COEF C_X (INF MASS)	AVERAGE ANGLE OF OSCILLATION, DEGREES	GENERAL APPLICATION
FLAT CIRCULAR			1.00	0.67 TO 0.70	0.75 TO 0.80	~1.7	±10 TO ±40	DESCENT, OBSOLETE
CONICAL			0.93 TO 0.95	0.70	0.75 TO 0.90	~1.8	±10 TO ±30	DESCENT, M < 0.5
BICONICAL			0.90 TO 0.95	0.70	0.75 TO 0.92	~1.8	±10 TO ±30	DESCENT, M < 0.5
TRICONICAL POLYCONICAL			0.90 TO 0.95	0.70	0.80 TO 0.96	~1.8	±10 TO ±20	DESCENT, M < 0.5
EXTENDED SKIRT 10% FLAT			0.86	0.66 TO 0.70	0.78 TO 0.87	~1.4	±10 TO ±15	DESCENT, M < 0.5
EXTENDED SKIRT 14.3% FULL			0.81 TO 0.85	0.66 TO 0.70	0.75 TO 0.90	~1.4	±10 TO ±15	DESCENT, M < 0.5

Drop-to-drop variations during test drops, from $n_{fill} \sim 5$ to $n_{fill} \sim 7$ – "facing the wind" deployments

(Figure by Desabrais & Johari [8])

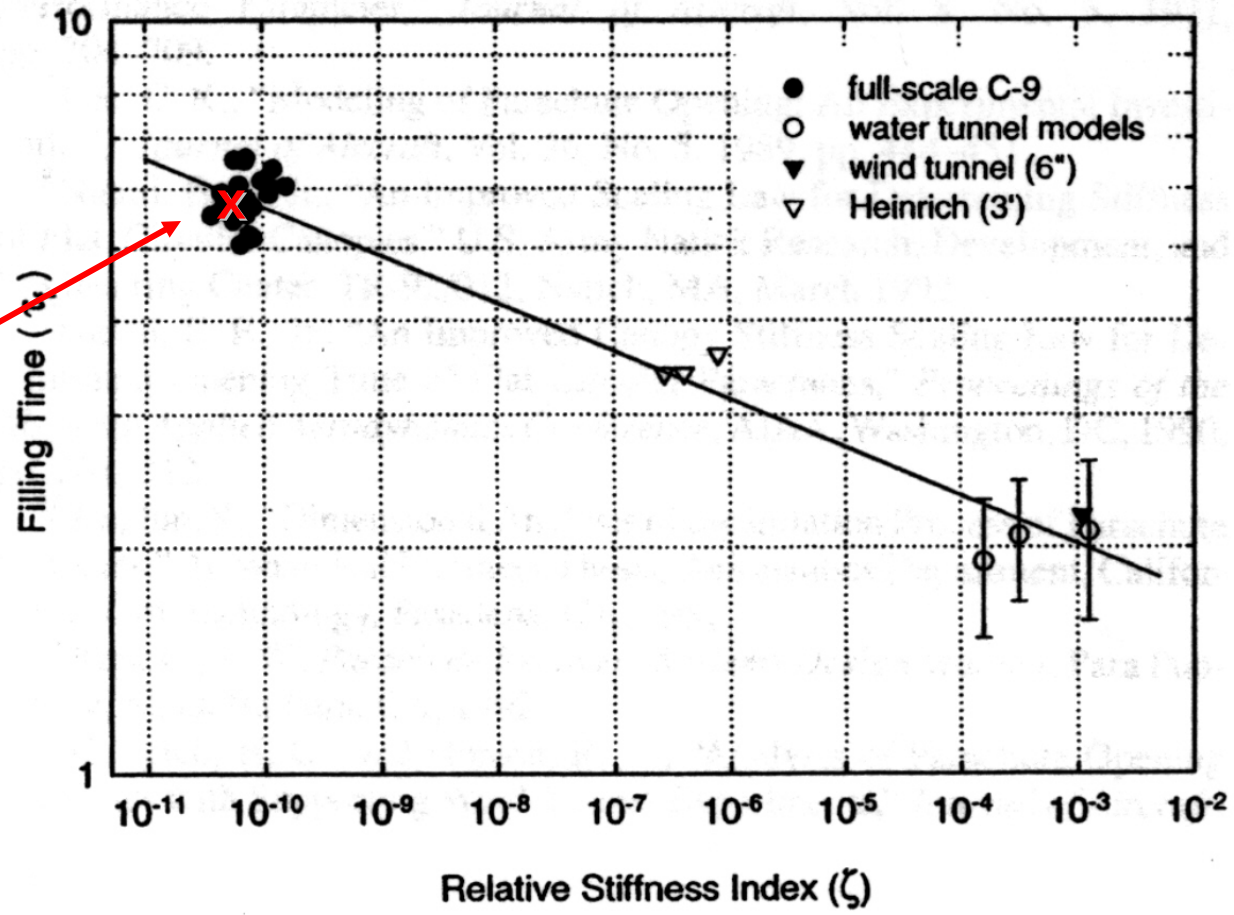


Figure 6.2 Choosing the non-dimensional inflation time for a C-9 type canopy (low-porosity hemispherical canopy)

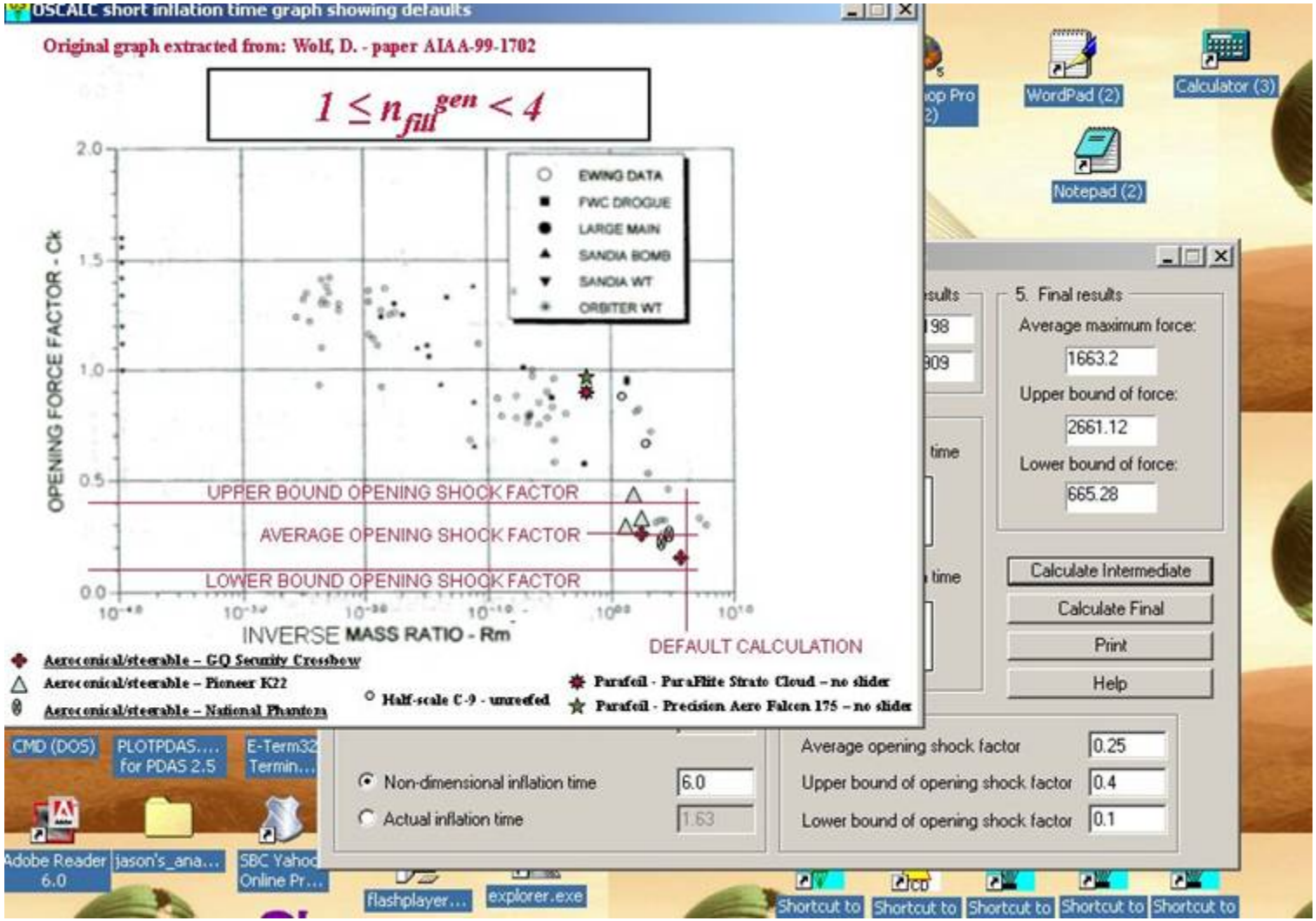


Figure 6.3 Chosen graph and C_k-values for the "Default" case

6.2 Permanently reefed high-porosity hemispherical canopy, deployed at high altitude and at low mass ratio

This example would be typical of 28ft-diameter ribbon-type canopy, reefed at 20%

ρ = 0.001 sl/ft^3 – deployment at 27,000ft MSL
m = 62.1 sl (corresponding to 2000 lbs on Earth)
$V_{stretch}$ = 300 ft/sec (typical if launched from cargo acft flying @ ~130KTSI)
$C_{D0}^{\ unreefed}$ = *0.38 – unreefed* - from reference [1] - see figure 6.4
S_0 = *616 ft^2* – calculated from $S_0 = \pi\ (28ft)^2/4$
$(SC_D)_{sd} = \varepsilon\ (SC_D)_{sd}|^{unreefed}$ = *0.27 x 0.38 x 616ft^2* = 63.2ft^2 - see figure 6.5

n_{fill} = *10* – See figure 6.6

OSCALC gives: R_m = 0.008 and n_{fill}^{gen} = 14.27
User should choose the "Long Inflation Time" graph
User would pick C_k = 1.25 with ~ 1.50 and ~ 1.00 as bounds

The result is F_{max} = 3465 lbs with 4158 lbs and 2772 lbs as bounds

Figure 6.4. Knacke [1] Tables 5-1 – 5-4.

TYPE	CONSTRUCTED SHAPE PLAN	CONSTRUCTED SHAPE PROFILE	$\frac{D_c}{D_o}$	INFLATED SHAPE $\frac{D_p}{D_o}$	DRAG COEF C_{D_o} RANGE	OPENING FORCE COEF C_X (INF MASS)	AVERAGE ANGLE OF OSCILLATION, DEGREES	GENERAL APPLICATION
FLAT (FIST) RIBBON			1.00	0.67	0.45 TO 0.50	~1.05	0 TO ±3	DROGUE, DESCENT, DECELERATION, OBSOLETE
CONICAL RIBBON			0.95 TO 0.97	0.70	0.50 TO 0.55	~1.05	0 TO ±3	DESCENT, DECELERATION, 0.1 < M < 2.0
CONICAL RIBBON (VARIED POROSITY)			0.97	0.70	0.55 TO 0.60	1.05 TO 1.30	0 TO ±3	DROGUE, DESCENT, DECELERATION, 0.1 < M < 2.0
RIBBON [1] (HEMISFLO)			0.62	0.62	0.30 [1] TO 0.46	1.00 TO 1.30	±2	SUPERSONIC, DROGUE, 1.0 < M < 3.0
RINGSLOT			1.00	0.67 TO 0.70	0.56 TO 0.65	~1.05	0 TO ±5	EXTRACTION, DECELERATION, 0.1 < M < 0.9
RINGSAIL			0.84	0.69	0.75 TO 0.85	~1.10	±5 TO ±10	DESCENT, M < 0.5
DISC–GAP–BAND			0.73	0.65	0.52 TO 0.58	~1.30	±10 TO ±15	DESCENT, M < 0.5

Figure 6.5

Using reference [1] to determine the drag area of a permanently reefed canopy. Note: using this curve for the ribbon chute of figures 6.4 (above) and 6.6 (below) may be an approximation, since the two chutes may not be exactly the same

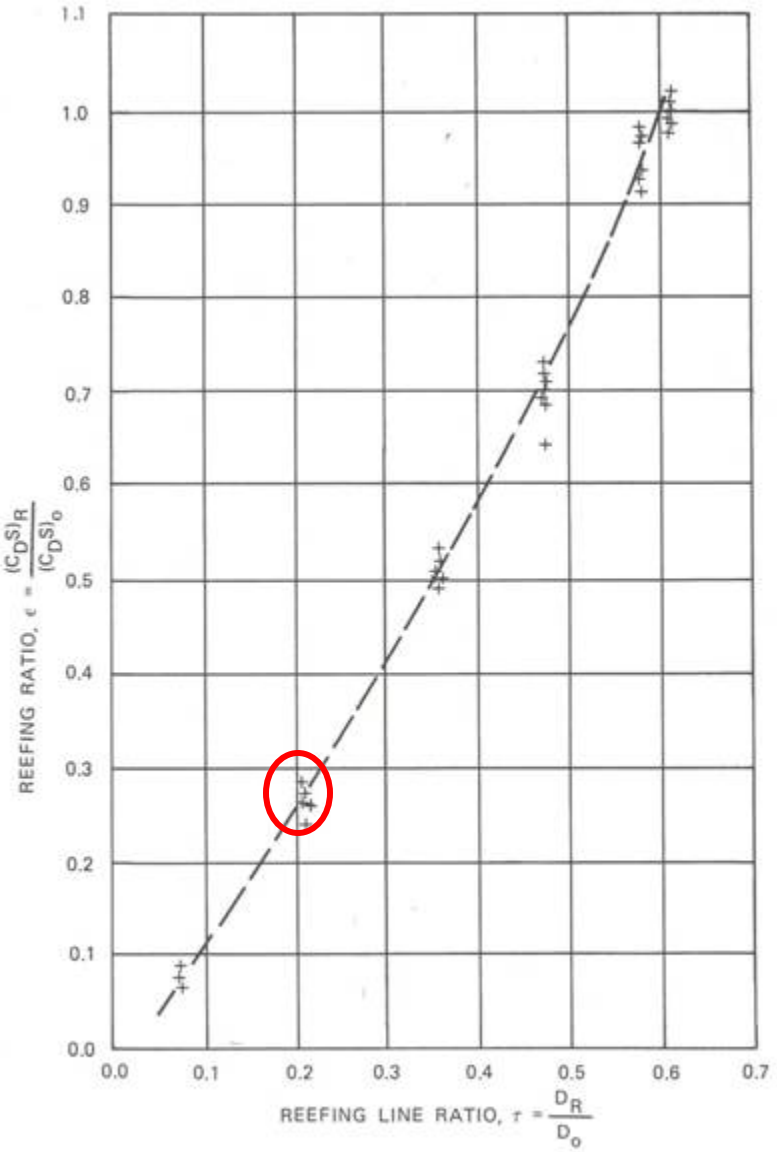


FIGURE 5-73. Reefing Ratio Versus Reefing-Line Ratio for 13.4-Foot-Diameter Conical Ribbon Parachute Tested in Aircraft Tow Test.

Figure 6.6 Choosing the non-dimensional filling time using reference [1].

Various Parachute Types.

Parachute type	Canopy fill constant, n		
	Reefed opening	Disreef opening	Unreefed opening
Solid flat circular	ID[a]	ID	8
Extended-skirt, 10%	16-18	4-5	10
Extended-skirt, full	16-18	7	12
Cross	ID	ID	11.7
Ribbon	10	6	14
Ringslot	ID	ID	14
Ringsail	7-8	2	7
Ribless guide surface	. . .	. . .	4-6

[a] ID = Insufficient data available for meaningful evaluation.

6.3 Un-reefed parafoil vs. dis-reefing parafoil

Part 1 - Consider the example of a 250ft^2 parafoil that is *not* equipped with a slider; this parafoil has no reefing whatsoever. In this example the values of S_0 is computed from the product of wing chord times span.

ρ = 0.002 sl/ft^3 – deployment at 5000ft MSL
m = 6.21 sl (corresponding to 200 lbs on Earth)
$V_{stretch}$ = 130 ft/sec
C_{D0} = *1.0* - during inflation the parafoil "looks" like a flat plate
S_0 = *250 ft^2*

n_{fill} = *2* – no reefing - see table, section 2

OSCALC gives: R_m = 1.27 and n_{fill}^{gen} = 1.13
User should choose the "Short Inflation Time" graph
User would pick C_k = 0.6 with ~ 0.9 and ~ 0.3 as bounds

The result is F_{max} = 2535 lbs with 3802 lbs and 1267 lbs as bounds

6.3 Un-reefed parafoil vs. dis-reefing parafoil

Part 2 - Consider the same 250ft^2 parafoil but equipped with a slider. Assume the same payload and deployment conditions. The only difference is the non-dimensional filling time which is increased (sliders do that).

ρ = 0.002 sl/ft^3 – deployment at 5000ft MSL
m = 6.21 sl (corresponding to 200 lbs on Earth)
$V_{stretch}$ = 130 ft/sec
C_{D0} = *1.0* - during inflation the parafoil "looks" like a flat plate
S_0 = *250 ft^2*

n_{fill} = *14* – See table, section 2 – old 1980's 7-cell design (they are in the n_{fill} ~ 15 - 25 range with the 1990's and 2000's designs)

OSCALC gives: R_m = 1.27 and n_{fill}^{gen} = 7.89
User should choose the "Long Inflation Time" graph
User would pick C_k = 0.2 with ~ 0.3 and ~ 0.1 as bounds

The result is F_{max} = 845 lbs with 1267 lbs and 422 lbs as bounds
QUITE A REDUCTION OF OPENING SHOCK!

6.4 Dis-reefing hemispherical canopy

Consider a 20%-reefed USAF C-9 canopy, which has been falling steadily in its reefed configuration, until reefing line cutter activation at 5000ft MSL

ρ = 0.002 sl/ft^3 – cutter activation at 5000ft MSL
m = 6.21 sl (corresponding to 200 lbs on Earth)
$V_{stretch} = V_{fall}$ = 40.3 ft/sec - using reefing drag area data of figure 6.7 to compute the fall rate prior to dis-reefing inflation
$C_{D0} = 0.75$ unreefed value (from reference [1]) - see figure 6.1
$S_0 = 616\ ft^2$ – calculated from $S_0 = \pi\ (28ft)^2/4$

$n_{fill} = 0.3 \times 6 = 1.8$ – See figure 6.8

OSCALC gives: R_m = 3.19 and n_{fill}^{gen} = 1.17
User should choose the "Short Inflation Time" graph
User would pick C_k = 0.3 with ~ 0.5 and ~ 0.15 as bounds

The result is F_{max} = 225 lbs with 375 lbs and 113 lbs as bounds

Figure 6.7

Using reference [1] to determine the drag area of a permanently reefed canopy.

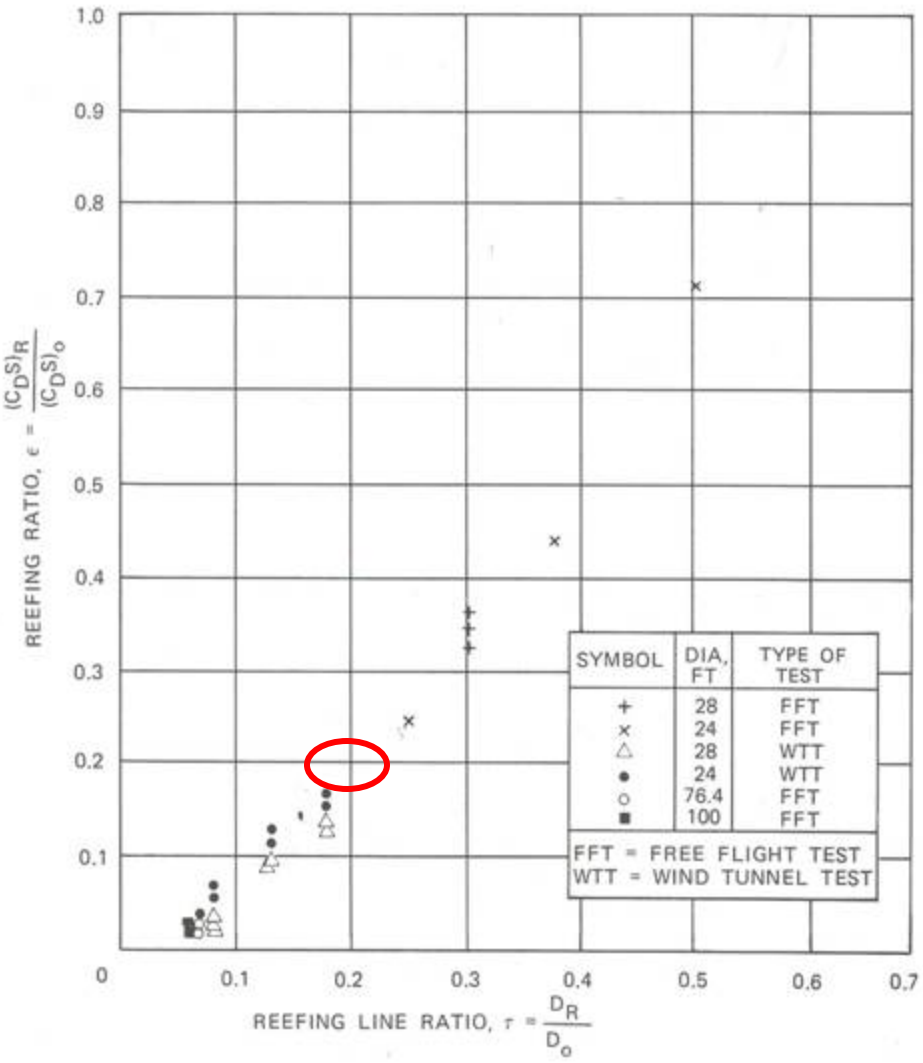


FIGURE 5-72. Reefing Ratio Versus Reefing-Line Ratio for Circular Flat, Conical, Triconical, and Extended-Skirt Parachutes.

Figure 6.8 Choosing the non-dimensional filling time.

Various Parachute Types.

Parachute type	Canopy fill constant, n		
	Reefed opening	Disreef opening	Unreefed opening
Solid flat circular	ID[a]	ID	8
Extended-skirt, 10%	16-18	4-5	10
Extended-skirt, full	16-18	7	12
Cross	ID	ID	11.7
Ribbon	10	6	14
Ringslot	ID	ID	14
Ringsail	7-8	2	7
Ribless guide surface	. . .	. . .	4-6

[a] ID = Insufficient data available for meaningful evaluation.

What to do?

Used simple estimate discussed in section 4 (limited to very low reefing ratios and low-porosity canopies)

$$\frac{n_{fill}^{disreef}}{n_{fill}^{unreef}} \sim 29 \frac{S_{init}^{mouth-unreef}}{D_0^2} \sim 0.29$$

6.5 Parachute cluster (unreefed)

Consider a cluster of four 100ft flat circular canopies.

$\rho = 0.002$ sl/ft^3 – deployment at 5000ft MSL
$m = 622$ sl (corresponding to 20,000 lbs on Earth)
$V_{stretch} = 200$ ft/sec
$C_{D0}^{\ single\ chute} = 0.75$ unreefed value (reference [1]) - see figure 6.1
$S_0 = 4 \times (\pi\ (100ft)^2/4) = 31{,}400ft^2$
We need the drag coefficient of the cluster - use figure 6.9:
$(SC_{D0})_{sd} = (0.83\ C_{D0}^{\ single})\ S_0 = 0.83 \times 0.75 \times 31{,}400ft^2 = 19{,}547ft^2$
So $C_{D0}^{\ 4\text{-}cluster} = 0.83\ C_{D0}^{\ single} = 0.62$ to be entered in OSCALC, together with S_0

$n_{fill}^{\ single} = 6$ – see figure 6.2
$n_{fill}^{\ 4\text{-}cluster} = 0.5 \times 6 = 3$ – See figure 6.10

OSCALC gives: $R_m = 8.73$ and $n_{fill}^{\ gen} = 2.15$
User should choose the "Short Inflation Time" graph
User would pick $C_k = 0.1$ with ~ 0.2 and ~ 0.05 as bounds

The result is $F_{max} = 77{,}872$lbs with 155,744lbs and 38,936lbs as bounds

Figure 6.9

Net drag coefficient of a parachute cluster (w/r to the drag coefficient of the member parachute making up the cluster

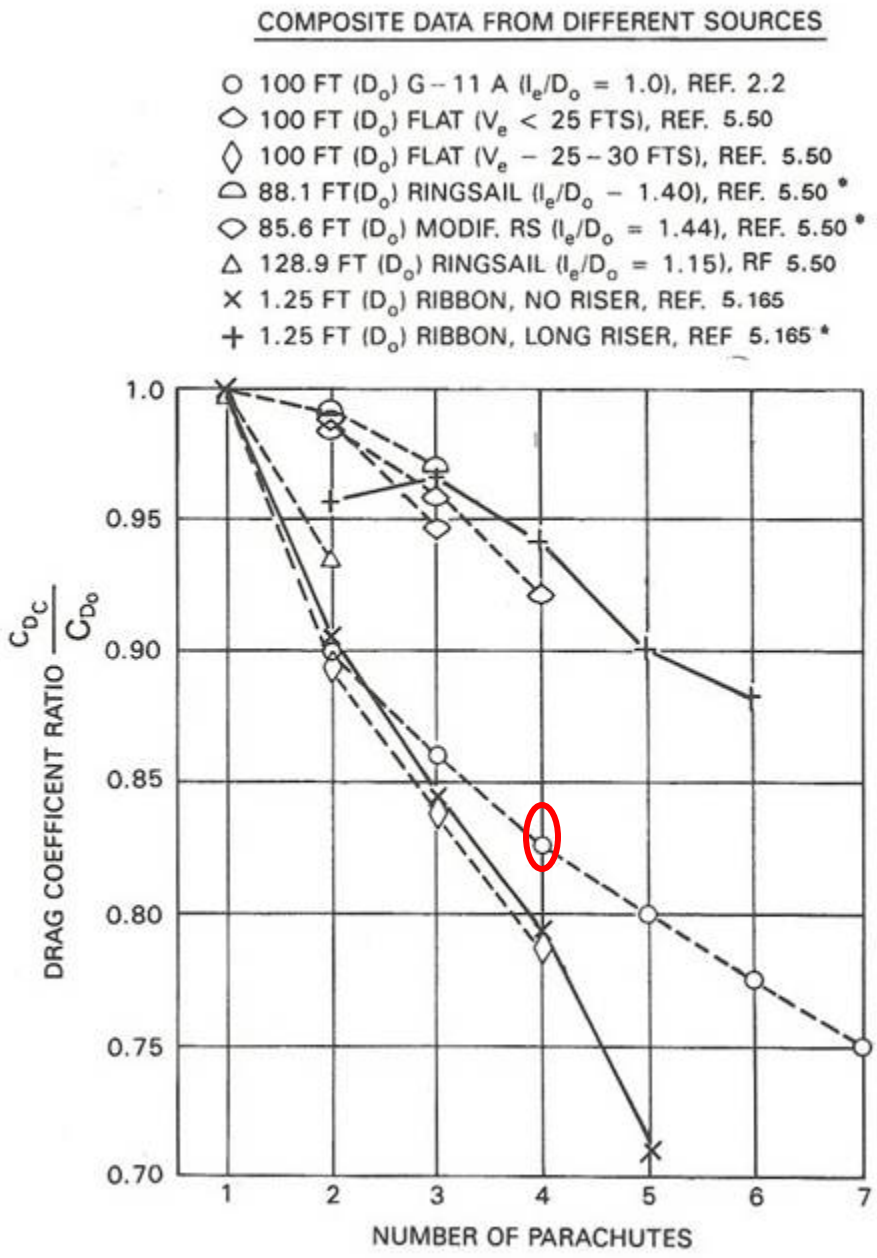


FIGURE 5-110. Drag Loss in Parachute Clusters.

Since the inflation time of the cluster ~
inflation time of each parachute of the cluster

then:

$$V_{stretch} t_{fill} = n_{fill}^{4-cluster} D_0^{4-cluster} = n_{fill}^{1-chute} D_0^{1-chute}$$

$$n_{fill}^{4-cluster} = n_{fill}^{1-chute} \sqrt{\frac{S_0^{1-chute}}{S_0^{4-cluster}}} = n_{fill}^{1-chute} \sqrt{\frac{1}{4}}$$

It is assumed here that all the parachutes of the cluster inflate at the same time and with the same inflation time – Warning - this is not always true! See Knacke's description of the "lead-lag" phenomenon

Figure 6.10 Assumptions used to compute the non-dimensional inflation time of a cluster.

6.6 What to do with a new design that is not documented in the World's database on inflation time and opening shock factor

- **The momentum-impulse theorem discussed in section 2 guarantees that the C_k data of the new parachute *will* duplicate the trend shown in the $C_k - R_m$ graphs, *as long as its generalized filling time is such that* $n_{fill}^{gen} \geq 1$**

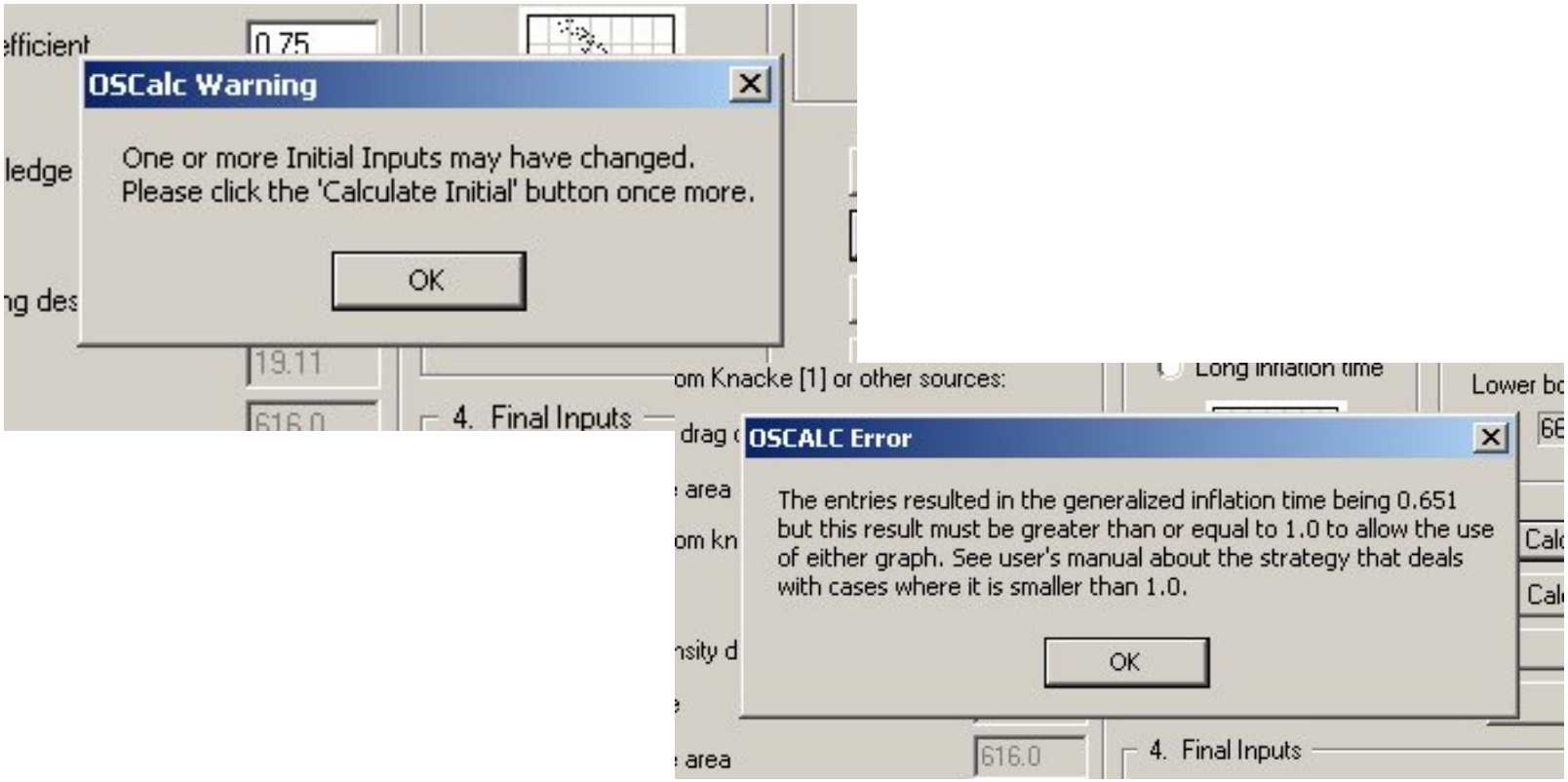
- **Coming up with an educated guess of the filling/inflation time may be possible – for example by using the values of documented parachute systems that are similar; or by guessing filling times values that could single out worse case scenarios (i.e. the largest realistic values of F_{max})**

- **Better filling time information will be obtained after the video of the "first flight" is analyzed**

7. OSCALC error/warning messages

7. OSCALC error/warning messages

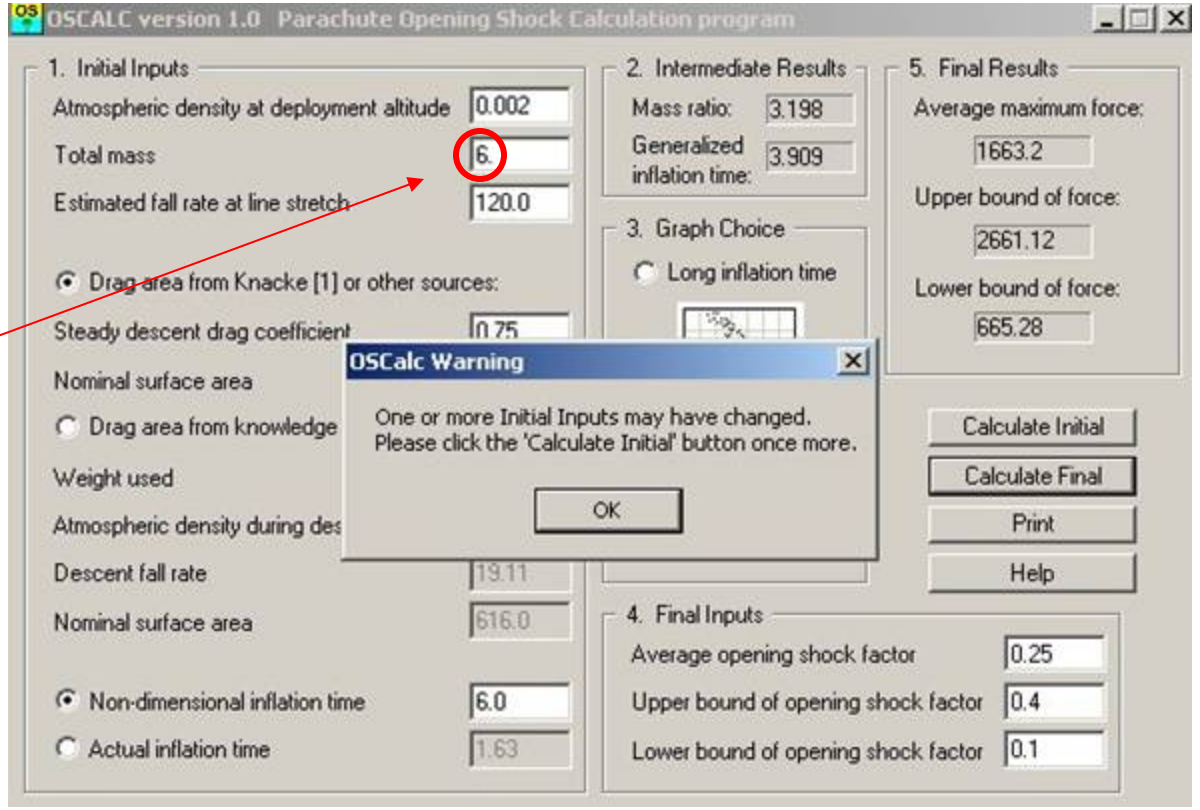
So far there are only two built-in warning/error messages:



OSCALC warning message

Warning: clicking the "Calculate Final" button before clicking the "Calculate Initial" button any time new input data is entered.

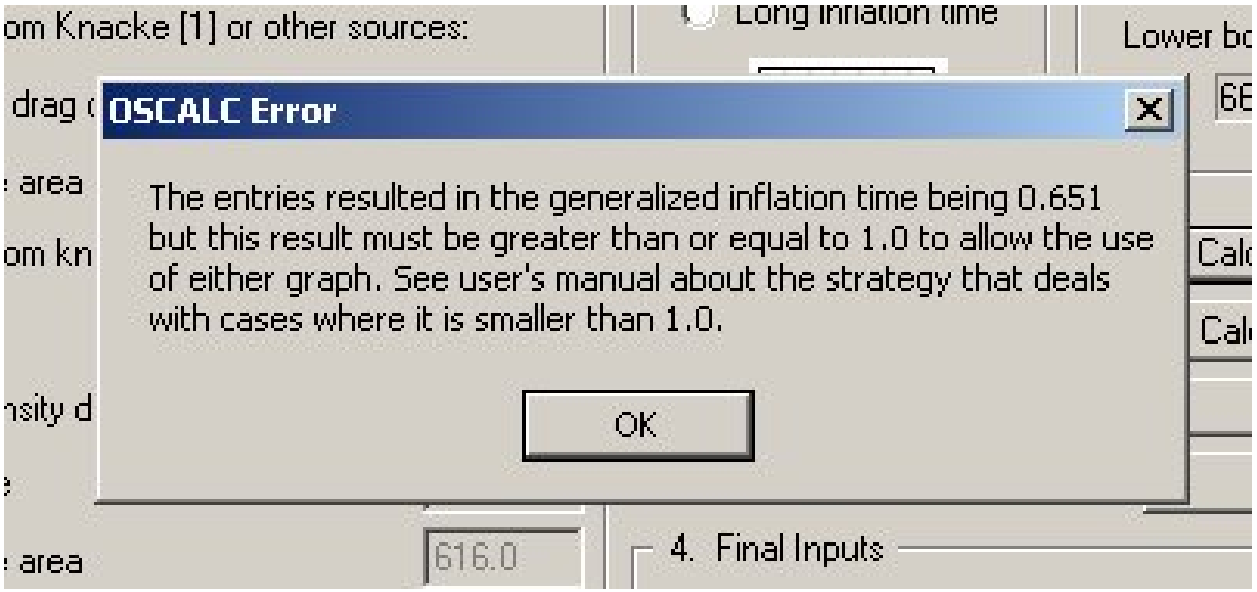
In this example, the mass was changed from "6.21" to "6", followed by the clicking of the "Calculate Final" button.



To clear the warning message: click "OK"; then click "Calculate Initial"

OSCALC error message

Error: Considering a case where the generalized filling time is less than unity.



See next page →

The case of very small filling time

- **This error message is triggered whenever $n_{fill}^{gen} < 1$. The problem is that neither graph apply to this case. If this value is not the result of a typing error, the user may consider using the Momentum-Impulse Theorem in tandem with OSCALC:**

  1) Run OSCALC for the same type of parachute and reefing, but with canopy diameters, atmospheric density and payload weight that is characterized by the same R_m *and a larger* n_{fill} ($\equiv n_{fill}^{alt}$), thereby obtaining a $C_k = C_k^{alt}$ *("alt" = "alternate")*

  2) Use the formula below to estimate the C_k of the actual system

$$C_k^{actual} = C_k^{alt} \frac{n_{fill}^{alt}}{n_{fill}^{actual}}$$

8. Concluding remarks

- **<u>This is the OSCALC-Version 1.0 family</u>. Future versions will provide updated $C_k - R_m$ graphs that include data points of parachute drops carried out more recently. These updates should close the gaps on the plots. This new data could also help create a new graph for the case $n_{fill}^{gen} < 1$**

- **Remember, the data scatter on the graphs is not due to lack of knowledge or measurement errors but rather to:**

  1) the integral Γ changing with flight angle (at the same value of R_m);

  2) n_{fill} being defined over a range of values (for each graph); and

  3) the drop-to-drop variations associated with both Γ and n_{fill}

  More details on all this can be found in reference [4]

- **OSCALC is not a true predictor of opening shock since it requires the use of t_{fill} (n_{fill}), which is an actual inflation performance variable**

- **OSCALC is not a design tool either, with regards to changing the dimensions of sub-components. For example, changing slider-size on a parafoil will be reflected in a change in t_{fill}. But even if the designer knew how to predict this change of t_{fill}, OSCALC may still give the same F_{max} if the new t_{fill} involves the same C_k-R_m graph. One way out of this problem, especially if t_{fill} is known, is to use the equation**

$$C_k = \frac{2}{R_m n_{fill}^{gen}} \Gamma$$ **shown on slide #20 and discussed in details in ref [4])**

- **Note that OSCALC can be of some use to the designer if the overall size and porosity of the canopy, payload weight, deployment altitude and/or pre-inflation fall speed are changed**

- **Design is where the detailed simulations of the drag force evolution would be most desirable; i.e. models sought to provide both F_{max} *and* t_{fill} *as* ***outputs****, depending on the actual design and constuction of the parachute, and on the actual drop conditions**

Will such models ever replace OSCALC?

- **Even with these detailed simulation tools being around, OSCALC will still be useful, in particular for calculating F_{max} sustained during inflation scenarios that are not covered by the detailed models, for example: malfunctions, mis-staged inflation or inflation sequences that begin with unusual canopy shapes**

Long live OSCALC!

Please send all questions to:

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